Investigation on the effect of temperature on the frequency of a vibrating cantilever system with a fixed length

Research question: How does temperature affect the natural frequency of a stainless steel cantilever beam at a fixed length?

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AI Acknowledgement

1. Essay Overview

The topic of mechanical vibration is covered in the IB Physics curriculum's simple harmonic motion (SHM) unit C.1, with the two prominent examples of a mass-spring system and a simple pendulum (Tsokos). The curriculum does not investigate more diverse mechanical vibrating systems, namely the cantilever beam. This is because the mathematical model of the cantilever system requires complex mathematics beyond the level of mathematics in IB physics. Instead, the cantilever beam is often used as an internal assessment topic, usually regarding the conspicuous length-frequency relationship.

This investigation goes beyond the typical length-based cantilever study by exploring the effect of temperature as an independent variable. Unlike length, whose effect on natural frequency follows directly from the frequency formula, the effect of temperature is indirect on natural frequency. The change in Young's modulus induced by change in temperature changes the natural frequency of the system, which is measurable and theoretically predictable (Al-hababi et al.; Sarhan and Al-Zubaidi).

The research question "How does temperature affect the natural frequency of a stainless steel cantilever beam at a fixed length?" holds significance because the temperature-dependence of structural vibration frequency is relevant to all industries (Xia et al.; Luo et al.) that operate mechanical systems across varying thermal environments. Bridge structures that experience seasonal temperature fluctuations, aerospace components that undergo extreme thermal variation during flight, and precision manufacturing equipment whose vibrational behavior must remain stable

across operating temperature are all examples of possible application of the temperature frequency relationship.

Thus, this essay aims to use mathematical modelling and experiments to investigate the effect of temperature on the natural frequency of a cantilever beam. A model is first derived in which the frequency falls linearly with temperature at a rate set by α_E, the temperature coefficient of Young's modulus; it is then tested by heating a clamped steel ruler in an oven and measuring its frequency.

2. Background Information

2.1 Cantilever Beams and Free Vibration

A cantilever beam is a beam that is rigidly fixed at one end while the other end remains free to move. When the free end is displaced from its equilibrium position, the elastic forces within the beam acting as a restoring force pull the end back towards equilibrium. This causes the beam to oscillate back and forth repeatedly, a motion known as free vibration ("Cantilever"). The rate at which this oscillation occurs is called the natural frequency, measured in hertz (Hz), and it is an intrinsic property of the system determined by its physical characteristics (Tsokos).

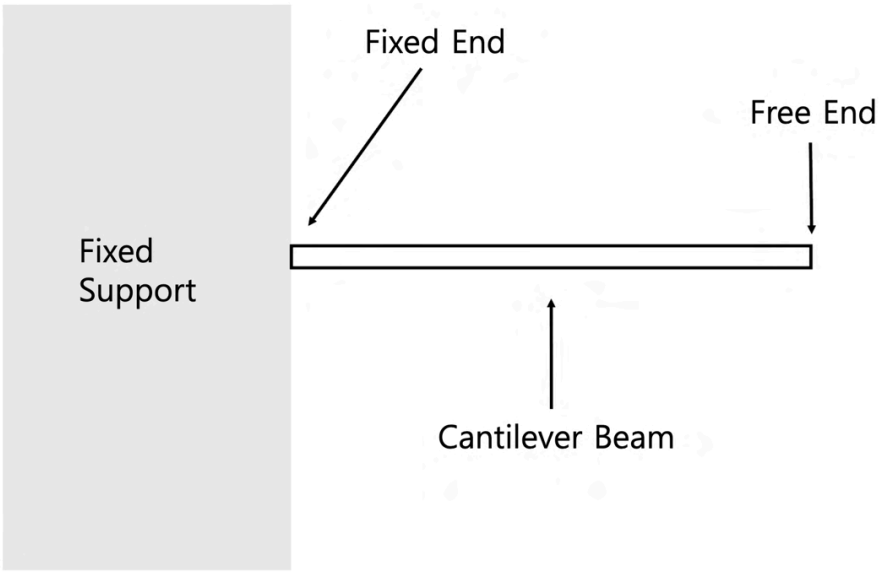


Figure 1. Diagram of a Cantilever Beam

Therefore, the natural frequency of the cantilever is governed by its bending stiffness and effective inertial mass, corresponding to the spring constant and mass terms in a mass-spring system respectively. These variables depend on the beam's geometry and material properties.

The bending stiffness of the beam is characterized by the product EI, where E is the Young's modulus of the material and I is the second moment of area of the cross-section, a geometric quantity of how the material is distributed relative to the bending axis ("Area Moment of Inertia").

2.2 Young's Modulus

Young's modulus is a fundamental material property that quantifies how resistant a material is to elastic deformation. It is defined as the ratio of tensile stress, the force applied per unit area, to strain, the fractional change in length that results, within the linear elastic region of the material ("Young's Modulus").

$$E = \frac{\sigma}{\varepsilon} = \frac{F/A}{\Delta L/L_0}$$

Equation 1. Definition of Young's Modulus (E) [Pa]. σ is tensile stress in [Pa] for force F [N] per area A [m²] and ε is tensile strain in [1] for change in length ΔL [m] over original length L_0 [m].

A material with a higher Young's modulus requires a greater force to produce the same amount of fractional deformation. At the atomic level, Young's modulus reflects the stiffness of the interatomic bonds within the material. When a material is deformed, atoms are displaced from their equilibrium positions, and the electrostatic restoring forces between them resist this displacement. The more force is required to displace the atoms, the more the atoms are resistant to stretches or compressions, giving the material its stiffness in a macroscopic scale (Isotta et al.). A material with stronger interatomic bonds, such as the metallic bonds in steel, therefore has a higher Young's modulus. This atomic origin of E is why temperature affects it, discussed in the following section.

Young's modulus is only constant within Hooke's region, a region where the stress-strain relationship is linear ("Young's Modulus"; Tsokos). In this investigation, the cantilever will go through very small oscillation amplitudes compared to its length, ensuring that deformations remain within the elastic regime.

2.3 Temperature Dependence of Young's Modulus

As temperature, a measure of the average kinetic energy of the particles, increases, atoms within the material gain kinetic energy and vibrate about their equilibrium

positions with greater amplitude. At higher amplitudes, the distance between atoms increases, making the interatomic restoring force weaker as electrostatic force is inversely proportional to distance squared. So, the effective stiffness of interatomic bonds decreases in higher temperature, decreasing Young's modulus as it is the macroscopic representation of the collective bond stiffnesses (Isotta et al.; Hirata and Shimonosono).

Since the natural frequency of the cantilever increases with the bending stiffness of the beam, like the spring constant in a mass-spring system, and the bending stiffness depends on Young's modulus, increasing the temperature of the cantilever makes it less stiff, which in turn causes it to oscillate slower. In short, temperature reduces Young's modulus, which reduces the natural frequency of the cantilever system, outlining the negative relationship between temperature and natural frequency (Iervolino et al.; Sarhan and Al-Zubaidi).

Additionally, increasing the temperature of a cantilever causes thermal expansion. Elongating the beam alone would lower the frequency, since f ∝ 1/L² in Equation 3, but the width and thickness expand by the same fraction, and the thickness enters the stiffness through I = bh³/12. For uniform expansion, every length is multiplied by the same factor s = 1 + α_thΔT, where α_th is the linear thermal expansion coefficient and ΔT is the temperature rise, while the mass stays the same. Following each term of Equation 3 through this scaling gives Equation R1.

$$L \to sL, \qquad I = \frac{bh^3}{12} \to s^4 I, \qquad A = bh \to s^2 A, \qquad \rho = \frac{m}{V} \to \frac{\rho}{s^3}$$

$$\Rightarrow \ \rho A \to \frac{\rho A}{s} \quad f = \frac{1.8751^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}} \ \to \ \frac{1}{s^2}\sqrt{s^4 \cdot s}\ f = s^{1/2} f \approx \left(1 + \frac{1}{2}\alpha_{th}\Delta T\right) f$$

Equation R1. Effect of uniform thermal expansion on Equation 3. The same result follows from Equation 4, since $3EI/L^3$ scales as s and the effective mass is unchanged.

For stainless steel, $\alpha_{th} \approx 1.7 \times 10^{-5}\ °C^{-1}$, so over the 113.6 °C range of this experiment expansion raises the frequency by $\frac{1}{2}(1.7 \times 10^{-5})(113.6) = +0.10\%$. The change in Young's modulus lowers it by $\frac{1}{2}\alpha_E \Delta T = \frac{1}{2}(4 \times 10^{-4})(113.6) = 2.3\%$ (Section 3.3). Thermal expansion is therefore about twenty times smaller than the change in Young's modulus and acts in the opposite direction, so it is considered negligible (Iervolino et al.).

3. Mathematical Model

3.1 Euler-Bernoulli Beam Theory

Euler-Bernoulli beam theory (EBBT) describes how slender beams bend and vibrate under transverse loading. The theory makes a few key assumptions: the beam's length is much greater than its thickness; the cross-sections of the beam remain flat and perpendicular to the neutral axis during bending, meaning that each section simply rotates with the beam rather than being distorted; the deflections are small relative to the beam's length; and the material is homogeneous, behaving the same way in every direction (Al-hababi et al.). In order to approximate these assumptions, a stainless steel ruler will be used as the cantilever with only a few millimeters displaced from rest.

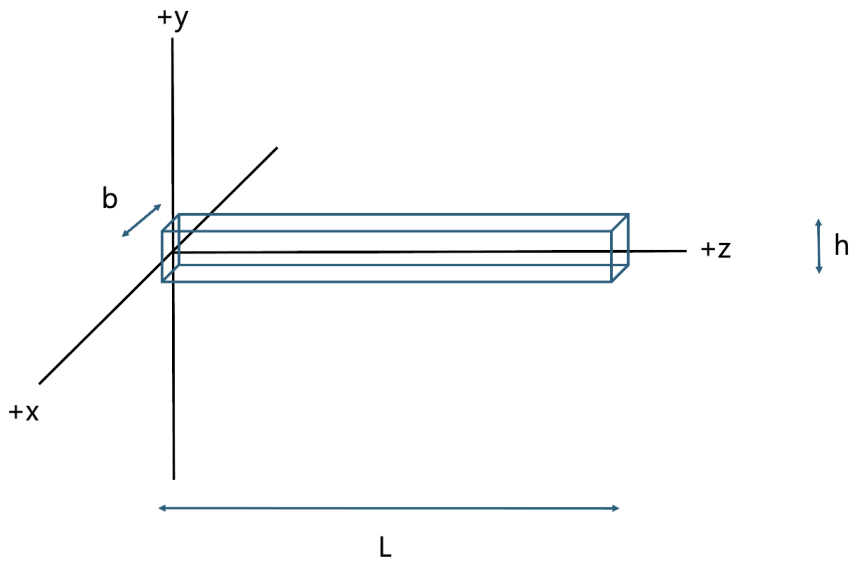


Figure 2. 3-dimensional diagram of a vibrating cantilever beam in an x, y, z plane. The neutral axis extends within the cantilever beam with the x-axis, and the transverse displacement is shown with the y-axis. The cantilever beam has width 'b', height 'h', and length 'L'.

Under these assumptions, the equation of motion for free, undamped transverse vibration of a uniform beam described by EBBT is:

$$EI\frac{\partial^4 w}{\partial x^4} + \rho A\frac{\partial^2 w}{\partial t^2} = 0$$

Equation 2. Euler-Bernoulli equation of motion for a uniform beam. w(x, t) is the transverse displacement (y-axis) of the beam at position x along its length and time t; ρ is the density of the beam material; and A is the cross-sectional area perpendicular to the axis of vibration.

The first term represents the internal bending resistance of the beam, and the second term represents the inertial resistance of its distributed mass, both as forces per unit

length [N m^{-1}]. This equation is the beam analogue of Newton's second law, as the elastic restoring force balances the inertial force at every point along the beam.

For a beam with a rectangular cross-section of width b and thickness h, the second moment of area is $I = bh^3/12$ ("Centroid"), derived in Appendix A.

3.2 Derivation of Natural Frequency

By solving the governing equation of EBBT using assumptions and boundary conditions given by that the cantilever beam vibrates at its first mode of vibration (one node at fixed end and one antinode at free length) without additional mass, the natural frequency of the cantilever beam can be derived as shown in Appendix B.

$$f = \frac{1.8751^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

Equation 3. Natural frequency of a cantilever beam derived from Euler-Bernoulli Beam Theory as shown in Appendix B.

However, as the experiment needs a method to measure the frequency of the oscillation, magnets at the tip of the ruler are added. These magnets act as additional point mass in addition to the mass of the ruler, needing a new equation for frequency. This is given through the Rayleigh method, which approximates the fundamental natural frequency of a vibrating system by approximating a non-point-mass oscillating object to an equivalent point-mass (Fletcher). Rayleigh method was used instead of EBBT with different boundary conditions as its solution became much more difficult to obtain, requiring complex mathematics.

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m_{eff}}} = \frac{1}{2\pi}\sqrt{\frac{3EI/L^3}{\frac{33}{140}\rho AL + m_{tip}}}$$

Equation 4. Natural frequency of the first mode with additional mass attached to the tip of the cantilever beam given through the Rayleigh method, derived in Appendix C. k_eff = 3EI/L³ is the stiffness of the tip and m_eff is the equivalent point mass at the tip.

Rayleigh method is still an accurate approximation for the math model, as it obtains similar results with EBBT. Rayleigh method becomes similar to EBBT when the mass of tip is zero, demonstrating it as an appropriate approximation shown in Appendix C.

With a tip mass, the fourth boundary condition of Appendix B changes: the shear force at x = L is no longer zero but equals the inertial force of the magnets. Solving EBBT with this condition gives the characteristic equation R2 in place of cos(βL)cosh(βL) + 1 = 0.

$$1 + cos(\beta L)cosh(\beta L) + \gamma\,\beta L[cos(\beta L)sinh(\beta L) - sin(\beta L)cosh(\beta L)] = 0, \qquad \gamma = \frac{m_{tip}}{\rho AL}$$

Equation R2. Exact characteristic equation for a cantilever with a point mass at the tip. β is defined as in Appendix B and γ is the ratio of tip mass to beam mass.

This equation has no closed-form solution, which is why Equation 4 is used. Solving it numerically for this ruler at L = 15 cm, where γ = 1.225 g / 22.69 g = 0.054 (Sample Calculations 1 and 5), gives $\beta_1 L = 1.785$, and the frequency is then $f = (\beta_1 L)^2/(2\pi L^2) \cdot \sqrt{(EI/\rho A)}$. Writing both results in the same form allows the coefficients to be compared directly.

$$f_{exact} = \frac{3.187}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}, \qquad f_{Rayleigh} = \frac{1}{2\pi L^2}\sqrt{\frac{3}{\frac{33}{140}+\gamma}}\sqrt{\frac{EI}{\rho A}} = \frac{3.218}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

Equation R3. Exact and Rayleigh frequencies for γ = 0.054, obtained by dividing the numerator and denominator of Equation 4 by ρAL.

The Rayleigh method therefore overestimates the frequency by 1.0% for this ruler, and by 1.5% when m_tip = 0 (√(3 × 140/33) = 3.568 against 1.8751² = 3.516). It always overestimates slightly because the static deflection shape is not exactly the vibration shape, and any approximate shape gives a frequency above the true one. Since the error is a constant factor, it cancels in the fractional gradient used to find α_E (Section 3.3).

3.3 Incorporating Temperature

As previously discussed in section 2.3, elastic modulus is dependent on temperature, with the approximate linear equation for moderate temperature range given by

$$E(T) = E_0 + E_0 \alpha_E \left(T - T_0\right)$$

Equation 5. Elastic Modulus (E) as a function of temperature (T). E_0 and T_0 respectively show elastic modulus and temperature at room temperature. α_E represents the approximated rate of change of elastic modulus for stainless steel within a moderate temperature region verified by different sources (Latella and Humphries; Al-hababi et al.).

The rate of change of the elastic modulus in respect to temperature is dependent on many different factors such as interatomic spring constant, thermal expansion coefficient, and specific heat capacity (Hirata and Shimonosono; Isotta et al.), but for moderate temperatures of 0°C to 300°C, the relationship can be approximated to a

negative linear relationship. For stainless steel, the coefficient is approximately α_E = -4 × 10^{-4} $°C^{-1}$ (Latella and Humphries).

By incorporating this equation to the frequency equation, the frequency of the cantilever beam as a function of temperature is given as

$$f = \frac{1}{2\pi}\sqrt{\frac{3I/L^3}{\frac{33}{140}\rho AL+m_{tip}}} \cdot \sqrt{E_0 + E_0\alpha_E\left(T - T_0\right)} = \frac{1}{2\pi}\sqrt{\frac{3I/L^3}{\frac{33}{140}\rho AL+m_{tip}}} \cdot \sqrt{E_0} \cdot \sqrt{1 + \alpha_E\left(T - T_0\right)}$$

Equations 6 and 7. Frequency as a function of temperature.

And for linear analysis, as $\alpha_E(T - T_0) << 1$, using Taylor expansion to the first term (Wathall et al. 63), the equation can be approximated to

$$f \approx \frac{1}{2\pi}\sqrt{\frac{3I/L^3}{\frac{33}{140}\rho AL+m_{tip}}} \cdot \sqrt{E_0} \cdot \left(1 + \frac{1}{2}\alpha_E T - \frac{1}{2}\alpha_E T_0\right)$$

Equation 8. Linear approximation of frequency as a function of temperature using Taylor expansion to the first term.

This gives a linear graph with the following key features:

$$\frac{df}{dT} = \frac{1}{2\pi}\sqrt{\frac{3I/L^3}{\frac{33}{140}\rho AL+m_{tip}}} \cdot \sqrt{E_0} \cdot \frac{1}{2}\alpha_E = \frac{1}{2}\alpha_E f_0, \qquad f_0 = \frac{1}{2\pi}\sqrt{\frac{3I/L^3}{\frac{33}{140}\rho AL+m_{tip}}} \cdot \sqrt{E_0}$$

Dividing the gradient by f_0 gives the fractional gradient, the fraction of the frequency lost per degree.

$$\frac{1}{f_0} \cdot \frac{df}{dT} = \frac{1}{2}\alpha_E$$

Equation R4. Fractional gradient of Equation 8.

Every constant in f_0 (I, L, ρ, A, m_{tip} and E_0) cancels, so the fractional gradient depends on α_E alone. For $\alpha_E = -4 \times 10^{-4}\ °C^{-1}$ the model predicts that the frequency falls by 0.020% per degree whatever the beam's dimensions, clamp or tip mass, provided these do not themselves change with temperature. This is the quantity the experiment is designed to test.

4. Variables

Independent Variable: Temperature of the steel ruler (°C)

The temperature of the ruler is the independent variable. It is varied across seven levels: room temperature (approximately 20°C), and six oven temperatures of approximately 40°C, 60°C, 80°C, 100°C, 120°C, and 140°C. At each temperature level, the ruler is allowed to reach thermal equilibrium with its surroundings before measurement. The ruler surface temperature is measured using an infrared thermometer (temperature gun) for all trials except room temperature, and a FLIR infrared camera for the room temperature trial, for reasons discussed in Section 5.1.

Dependent Variable: Natural frequency of the cantilever system (Hz)

The natural frequency is the dependent variable. It is measured using a Hall-effect magnetic field sensor placed near the tip magnet of the ruler. When the free end of the ruler is displaced and released, the oscillating magnet produces a periodically varying magnetic field, which the sensor records at a sampling rate of 1000 readings per second over a 20-second window. The natural frequency is then determined by finding the main frequency of the recorded oscillations.

Controlled Variables

Variable controlled	Reason for control	Means of control
Free length of ruler (L = 15 cm)	Any unintended change in free length would shift the natural frequency independently of temperature, altering the result.	Two wooden blocks sandwich the ruler at exactly the 15 cm mark, held in place by two metal G-clamps tightened firmly.
Mass and position of tip magnets	The tip magnets contribute their full mass to the effective inertia. Any change in mass or position would alter the frequency independently of temperature.	The same magnets are fixed at the tip of the ruler throughout all trials. Each magnet is positioned so that half of its width overhangs the edge of the ruler, centred along the width of the ruler, approximating a point mass at the tip.
Clamping arrangement	The rigidity of the clamp sets the boundary condition at the fixed end, so a looser clamp behaves like a longer beam and lowers the frequency.	The same wooden blocks and G-clamps are used for every temperature trial.
Sampling rate and duration of frequency measurement	A consistent sampling rate and recording window ensures that the frequency resolution and signal quality are identical across all trials, making readings directly comparable.	The magnetic field sensor records at 1000 readings per second for 20 seconds in every trial.
Time for ruler to reach goal temperature	Without sufficient time, the temperature of the ruler might vary within or not reach the goal temperature.	Around 5 to 10 minutes per temperature increment is used for temperature to reach equilibrium in the oven.

5. Methodology

5.1 Measurement of Constants

Before conducting the main experiment, the measurement of constants should be done. Especially, the constant of Young's modulus at room temperature (E_0) needs to be figured out by conducting a preliminary experiment of measuring natural frequency with variable length in room temperature.

For this preliminary experiment, the ruler is placed horizontally on the edge of a table with the desired free length hanging off the edge. The clamping is achieved by stacking multiple textbooks on top of the ruler at the chosen length point, applying sufficient pressure to prevent displacement at the fixed end. The magnetic field sensor is placed next to the tip magnet; the free end is displaced and released; and the frequency is recorded at the same sampling rate of 1000 readings per second over 20 seconds. This is repeated for several different free lengths. Since this experiment is conducted at room temperature, the FLIR infrared camera is used to record the ruler surface temperature, as the infrared thermometer is not accurate below approximately 50°C.

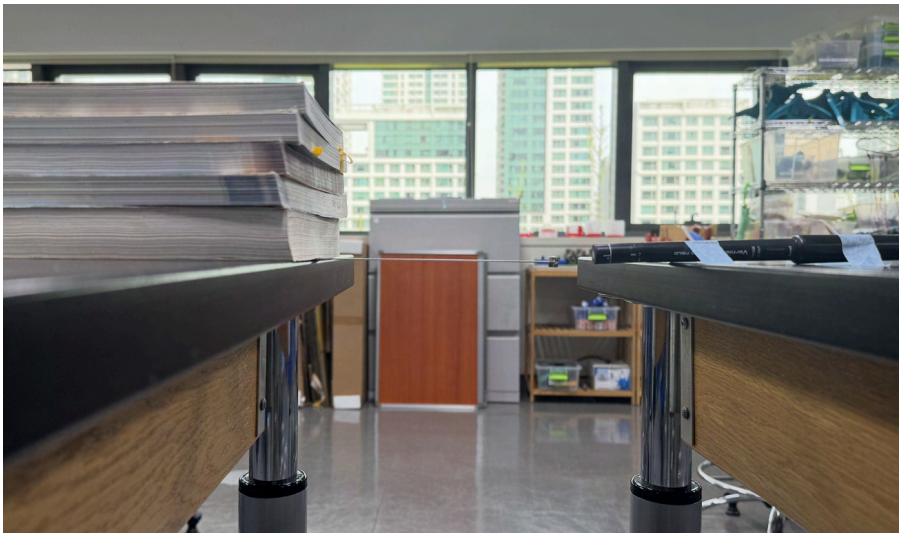


Figure 3. Experimental setup for measuring elastic modulus at room temperature by measuring different frequencies at different lengths.

Quantity measured	Value	Uncertainty (±)
Mass of ruler	49.75 g	0.01 g
Mass of magnets (eight in total)	4.90 g	0.01 g
Thickness of ruler	0.46 mm	0.005 mm
Width of ruler	26 mm	0.5 mm
Height of magnet	2.68 mm	0.005 mm
Diameter of magnet	5.59 mm	0.005 mm

Quantity measured Ruler empty circle diameter	Value 6.25 mm	Uncertainty (±) 0.01 mm
Height of the curvature part of ruler	4.33 mm	0.01 mm
Length of ruler before curvature	327.0 mm	0.05 mm

Table 1. Raw data table for results of constant measurements.

Constants from the derived math model can now be determined. From the data table above, constants in the math model can be defined in SI units.

$$f = \frac{1}{2\pi}\sqrt{\frac{3I/L^3}{\frac{33}{140}\rho AL + m_{tip}}} \cdot \sqrt{E_0} \cdot \left(1 + \frac{1}{2}\alpha_E T - \frac{1}{2}\alpha_E T_0\right);$$

$$L = 0.15\ m\ (for\ different\ temperature\ and\ fixed\ length)$$

$$I = \frac{bh^3}{12} = \frac{26\times10^{-3}\cdot\left(0.46\times10^{-3}\right)^3}{12} = 0.210 \times 10^{-12}\ m^4, \qquad A = bh = 11.96 \times 10^{-6}\ m^2$$

$$m_{tip} = 2 \times 0.6125\ g = 1.225 \times 10^{-3}\ kg, \qquad \rho = \frac{m}{V} = \frac{49.75\times10^{-3}}{3932.0846\times10^{-9}} = 12650\ kg\ m^{-3}$$

Equation 9. The value of constants in SI units. Derivation of density in Appendix D. Only two of the eight magnets (0.6125 g each) are on the ruler.

Length (cm) 10	Average frequency (Hz) 43.6531	± combined (Hz) 0.2074	σ (Hz) 0.3843	n trials 4	x (m^{-3} kg^{-1}) 208686.8242	y = f^2 (Hz^2) 1905.59314
15	22.4741	0.0087	0	1	45061.9917	505.0851708
20	13.337	0.0754	0.1257	3	14954.4093	177.875569
25	8.9056	0.0309	0.0167	3	6310.2845	79.30971136
30	6.2698	0.0366	0.0653	9	3105.6728	39.31039204

Table 2. Processed data table for frequency with various lengths at room temperature. Raw data and the method of frequency extraction are in Appendix E (Table E2).

From the math model, E_0 can be derived. The horizontal axis (x) is set equal to L^{-3} / ((33/140)ρAL + m_tip) and the vertical axis (y) to f^2, so that the slope dy/dx equals $3IE_0$ / $4\pi^2$.

$$f^2 = \frac{3IE_0}{4\pi^2} \cdot \frac{L^{-3}}{\frac{33}{140}\rho AL + m_{tip}} \quad \Rightarrow \quad y = \frac{3IE_0}{4\pi^2}\, x \quad \Rightarrow \quad E_0 = \frac{4\pi^2}{3I} \cdot \frac{dy}{dx}$$

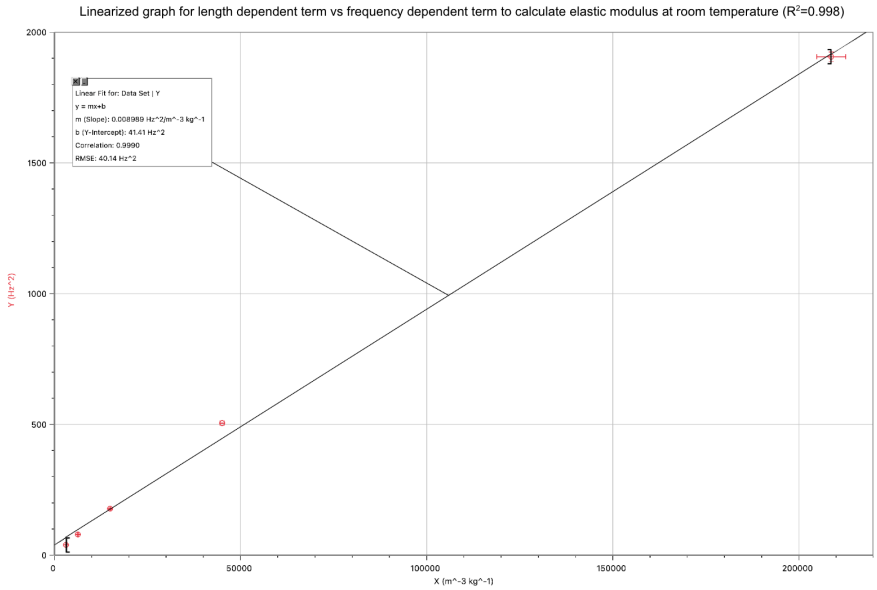


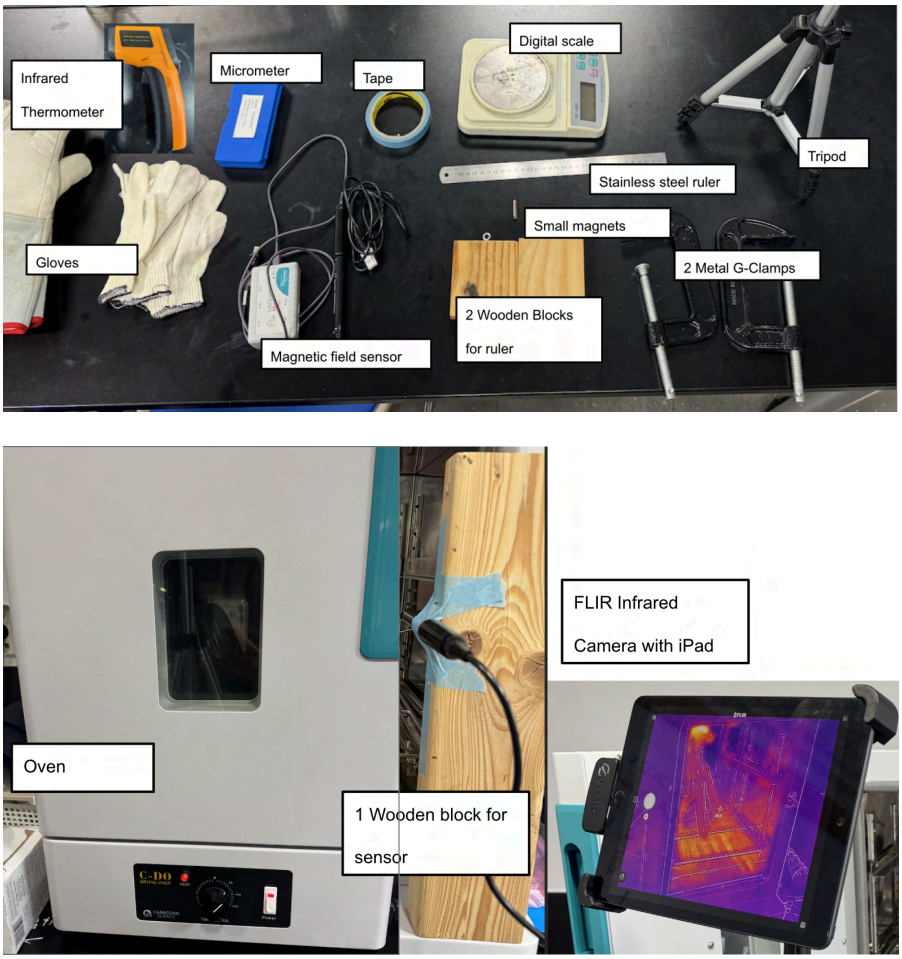
Figure 4. Linearized graph for length dependent term vs frequency dependent term to calculate elastic modulus at room temperature.

So, using the slope of the graph above, the elastic modulus at room temperature E_0 = (5.61 ± 0.43) × 10^{11} Pa (Sample Calculation 10). This value is revisited in Section 7.

This value is already about 2.8 times the accepted 193 to 200 GPa for stainless steel ("Young's Modulus"), and the points of Figure 4 do not lie on a straight line through the origin: the short lengths fall below the line drawn through the origin and the long ones (at 10 cm, f^2 is 1906 Hz^2 against 2641 Hz^2 on the line through the 30 cm point). Both point to the textbook clamp and the measured thickness rather than to the model, and

they are examined in Section 7.3. Because α_E comes from the fractional gradient (Equation R4), the temperature analysis does not depend on this value of E_0.

5.2 Experiment Setup



Figures 5 and 6. Photos of materials.

Materials List:

- 1 Stainless steel ruler (30 cm+)
- 2 Wooden blocks for fixing the ruler
- 1 Wooden block for attaching the magnetic field sensor
- 2 Metal G-clamps

- 2 Small magnets
- 1 Hall-effect magnetic field sensor
- 1 Tape
- 1 Oven (~140 °C)
- 1 Digital balance scale (g ±0.01)
- 1 Micrometer (mm ±0.01)
- 2 Gloves
- 1 FLIR infrared camera (°C ±0.1) with iPad
- 1 Infrared thermometer (°C ±0.1)

The main experimental setup consists of three components: the cantilever assembly, the temperature control system, and the frequency measurement system.



Figure 7. Experimental setup for differing temperatures in an oven.

For the cantilever assembly, the steel ruler is sandwiched between two wooden blocks at the 15 cm mark from the free end. Two metal G-clamps are tightened firmly around the outer faces of the blocks, applying sufficient pressure to prevent any displacement or rotation of the ruler at the clamp point. A small neodymium magnet is attached to

each face of the ruler at the free tip, positioned so that it approximates a point mass at the tip of the ruler.

For the temperature control system, the cantilever assembly is placed inside an oven at an angle to fit within the oven. The oven is set to the target temperature and allowed to reach steady state before measurement, ensuring that the temperature inside the oven and the ruler surface are at equilibrium. For the room temperature trial, the assembly is held on a flat surface table without heating.

For the frequency measurement system, a Hall-effect magnetic field sensor is positioned near the tip magnet of the ruler by being attached to a wooden block using tape. The distance between the sensor and the magnet varies slightly between trials due to replacing it after opening the oven door, but since only the frequency of the signal is important for analysis, this variation can be ignored.

5.3 Experiment Procedure and Safety

- Assemble the cantilever: sandwich the ruler between the two wooden blocks at the 15 cm mark and tighten the two G-clamps firmly.
- Confirm that the tip magnets are attached correctly at the free end.
- Place the cantilever assembly inside the oven at an angle.
- Set the oven to the first target temperature (40°C) and allow it to reach steady state for at least 5 minutes.
- Open the oven door and immediately position the magnetic field sensor near the tip magnet.

- Begin recording the magnetic field signal at 1000 readings per second in a 20 seconds time frame.
- Displace the free tip of the ruler by a small amount and release.
- Repeat step 7 for at least 5 trials within the 20 seconds time frame.
- Point the infrared thermometer at the ruler surface and record the temperature reading immediately after the frequency measurement.
- Close the oven door and increase to the next target temperature (60°C). Wait at least 5 to 10 minutes for equilibration.
- Repeat steps 5 to 10 for target temperatures of approximately 60°C, 80°C, 100°C, 120°C, and 140°C.
- Repeat steps 5 to 9 at room temperature, using the FLIR camera instead of the infrared thermometer for the temperature measurement.

Safety Precautions:

The experiment includes being exposed to high temperatures due to the air flowing out of the oven and displacing the tip of the ruler by hand. Due to this safety concern, it is crucial to wear 2+ layers of gloves for thermal insulation and to not get too close to the oven.

6. Data Collection and Processing

6.1 Qualitative Observations

Several qualitative observations that are relevant to understanding potential sources of error in the data were made.

The most notable observation was that the oven tray vibrated visibly when vibrating the ruler inside the oven. This is because the cantilever assembly was resting against the oven tray, allowing mechanical vibrations to propagate into the tray. However, since the frequency analysis only extracts the dominant oscillation frequency of the ruler and the ruler's natural frequency is independent from the resonance of the oven tray, this should have negligible effect on the frequency readings. In the data, the existence of a low frequency (below 8 Hz) confirms this noise, but since the natural frequency of the ruler is much larger than that, it does not affect the data.

A significant practical issue was the measurement of the ruler surface temperature inside the oven. The FLIR infrared camera, which was successfully used at room temperature, became unable to accurately measure the temperature of the ruler surface when the ruler was inside the oven. The camera readings were distorted by the oven interior thermal environment, reading temperature at least 10 degrees Celsius different from the set temperature. So, the infrared thermometer was used for all oven trials instead. However, the temperature gun has a specified accurate measurement range beginning at approximately 50°C. As the lowest oven trial temperature of approximately 40°C falls below this range, it should be treated with caution of error in the analysis. In addition, the numerical reading of the infrared thermometer was observed to fluctuate significantly during measurement instead of stabilizing to a fixed value, introducing unrecorded uncertainty to the measured temperature at each trial.

Finally, opening the oven door to perform frequency measurements resulted in the setup being briefly exposed to room temperature air, which would slightly cool the ruler surface before the measurement was completed. This effect is considered to be

minimal, since the temperature was recorded immediately after the frequency measurement: the recorded temperature represents the ruler's temperature at the time of measurement, not the setpoint of the oven.

The sensor had to be repositioned each time the door was opened, so its distance from the magnet changed between recordings. This changed the size of the recorded signal but not its period: recordings with large and small signals gave frequencies within the same range at each temperature (Appendix E, Table E1), which confirms that only the timing of the signal matters for the analysis.

A recording at about 2 °C, made after cooling the assembly, is listed in Appendix E but not used in the analysis, since the ruler's temperature during the flicks could not be measured reliably.

The ruler was displaced by hand, so the initial displacement varied between flicks, and so did how long each flick lasted before it decayed into the noise: accepted segments ranged from 0.8 s to 17.2 s (Appendix E). The effect of this on the measured frequency is examined in Section 7.4.

6.2 Uncertainty Analysis

Table 3 lists the uncertainty of every measured quantity used in the calculations; the magnet height and diameter in Table 1 were not used and are omitted. Uncertainties were propagated by adding percentage uncertainties for products, quotients and powers, and absolute uncertainties for sums (Equation 10; Tsokos). Sample Calculations 1 to 12 show one example of each propagation; Table 4 summarises them.

Quantity	Value	Absolute uncertainty	Percentage uncertainty
Mass of ruler, m_r	49.75 g	±0.01 g	0.020%
Mass of eight magnets	4.90 g	±0.01 g	0.20%
Thickness of ruler, h	0.46 mm	±0.005 mm	1.09%
Width of ruler, b	26 mm	±0.5 mm	1.92%
Length of ruler before curvature, l	327.0 mm	±0.05 mm	0.015%
Ruler empty circle diameter, d	6.25 mm	±0.01 mm	0.16%
Height of the curvature, h_c	4.33 mm	±0.01 mm	0.23%
Free length, L	150 mm	±0.5 mm	0.33%
Temperature, T	24.8 to 138.4 °C	±0.1 °C	0.07 to 0.40%

Table 3. Uncertainties of the measured quantities used in the calculations.

$$Q = k\,x^{a}y^{b} \;\Rightarrow\; \frac{\Delta Q}{Q} = |a|\frac{\Delta x}{x} + |b|\frac{\Delta y}{y}, \qquad Q = x \pm y \;\Rightarrow\; \Delta Q = \Delta x + \Delta y$$

Equation 10. Propagation of uncertainty for products and powers, and for sums and differences. k is a constant with no uncertainty and a, b are powers.

$$m_{tip} = 2 \times \frac{4.90\ g}{8} = 1.225\ g, \qquad \Delta m_{tip} = 2 \times \frac{0.01\ g}{8} = 0.0025\ g\ (0.20\%)$$

Sample Calculation 1. Tip mass, from the mass of eight magnets.

$$\frac{\Delta I}{I} = \frac{\Delta b}{b} + 3\frac{\Delta h}{h} = \frac{0.5}{26} + 3\left(\frac{0.005}{0.46}\right) = 1.92\% + 3.26\% = 5.18\% \Rightarrow I = (2.109 \pm 0.109) \times 10^{-13}\ m^4$$

Sample Calculation 2. Second moment of area; the cubed thickness dominates.

$$\Delta A_{rectangle} = 8502\ mm^2 \times \left(\frac{0.5}{26} + \frac{0.05}{327.0}\right) = 164.8\ mm^2$$

$$\Delta A_{circle} = 30.68\ mm^2 \times 2\left(\frac{0.01}{6.25}\right) = 0.10\ mm^2$$

$$\Delta A_{curvature} \approx 76.69\ mm^2 \times \left(\frac{0.5}{26} + \frac{0.01}{4.33}\right) = 1.65\ mm^2$$

$$\Delta A_{total} = 164.8 + 1.65 + 0.10 = 166.6\ mm^2\ (1.95\%)$$

Sample Calculation 3. Total face area of the ruler (Appendix D), a sum of three areas. The curved segment is treated as proportional to b · h_c.

$$\frac{\Delta\rho}{\rho} = \frac{\Delta m_r}{m_r} + \frac{\Delta A_{total}}{A_{total}} + \frac{\Delta h}{h} = 0.02\% + 1.95\% + 1.09\% = 3.06\% \quad \Rightarrow \quad \rho = 12,650 \pm 390\ kg\ m^{-3}$$

Sample Calculation 4. Density of the ruler, ρ = m_r / (A_total · h).

The effective mass is the denominator of Equation 4, so its uncertainty is propagated from ρ (Sample Calculation 4), A = bh (Equation 9) and L, as in Sample Calculations 5 and 6.

$$\frac{\Delta A}{A} = \frac{\Delta b}{b} + \frac{\Delta h}{h} = 1.92\% + 1.09\% = 3.01\% \quad \frac{\Delta(\rho AL)}{\rho AL} = \frac{\Delta\rho}{\rho} + \frac{\Delta A}{A} + \frac{\Delta L}{L}$$

$$= 3.06\% + 3.01\% + \frac{0.5}{150} = 6.40\% \quad \rho AL = 12650 \times \left(11.96 \times 10^{-6}\right) \times 0.150$$

$$= 22.69 \times 10^{-3}\ kg \quad \Rightarrow \quad \rho AL = 22.69 \pm 1.45\ g$$

Sample Calculation 5. The beam term ρAL in the denominator of Equation 4 (L = 0.150 m), with ρ from Sample Calculation 4 and A = bh from Equation 9. Treating ρ and A as independent counts b and h twice, so this uncertainty is an upper bound.

$$m_{eff} = \frac{33}{140}\rho AL + m_{tip} = \frac{33}{140}(22.69\ g) + 1.225\ g = 6.574\ g,$$

$$\Delta m_{eff} = \frac{33}{140}(1.45\ g) + 0.0025\ g = 0.345\ g\ (5.25\%)$$

Sample Calculation 6. Effective mass, the denominator of Equation 4: a constant multiple of ρAL plus m_tip, so the absolute uncertainties add.

Each trial's frequency uncertainty is one tenth of the raw FFT bin width (Section 6.3, Equation 11), a conservative estimate, since tests on synthetic signals gave errors

below 0.003 Hz (Appendix F). Each mean combines this with the scatter between trials (Equation 12).

$$\varepsilon = \frac{\Delta f_{raw}}{10}, \qquad \Delta f_{raw} = \frac{1}{t_{segment}}$$

Equation 11. Per-trial instrumental uncertainty in frequency, where t_segment is the duration of the flick being analysed.

$$\delta f = \sqrt{\left(\frac{\sigma}{\sqrt{n}}\right)^2 + \varepsilon^{-2}}$$

Equation 12. Combined uncertainty in the mean frequency, where σ is the standard deviation across n accepted trials and ε̄ is the mean per-trial uncertainty.

Equation 11 comes from how an FFT works. A recording lasting t seconds can only be split into frequencies spaced 1/t apart, so a 1.76 s flick gives bins 0.568 Hz wide; the shorter the flick, the coarser the bins. Zero-padding and parabolic interpolation (Section 6.3) locate the peak between bins, so the true uncertainty is a small fraction of a bin. Taking one tenth of a bin is a cautious choice: on synthetic signals of known frequency, the method's error was under 0.003 Hz, at least 20 times smaller than ε (Appendix F).

Equation 12 combines two separate sources of uncertainty in a mean. The first, σ/√n, is how much the mean could move because individual flicks scattered; it shrinks as more flicks are averaged. The second, ε̄, is the measurement precision of each flick, which averaging does not remove because every flick is measured the same way. The two are independent, so they are added in quadrature, the square root of the sum of squares, rather than directly.

$$\Delta f_{raw} = \frac{1000\ Hz}{1760} = 0.568\ Hz \quad \Rightarrow \quad \varepsilon = 0.057\ Hz$$

Sample Calculation 7. First trial at 82.1 °C (1760 samples at 1000 readings per second, so t_segment = 1.760 s).

$$\delta f = \sqrt{\left(\frac{0.0413}{\sqrt{8}}\right)^2 + 0.0681^2} = \sqrt{0.0146^2 + 0.0681^2} = 0.0696\ Hz$$

Sample Calculation 8. Mean frequency at 82.1 °C (Table 5), where 0.0681 Hz is the mean of the eight per-trial ε values.

$$\frac{\Delta(f^2)}{f^2} = 2\frac{\delta f}{f} = 2\left(\frac{0.2074}{43.6531}\right) = 0.95\% \quad \Rightarrow \quad y = 1905.6 \pm 18.1\ Hz^2$$

Sample Calculation 9. Vertical axis of Figure 4 at L = 10 cm (Table 2).

$$E_0 = \frac{4\pi^2}{3I} \cdot \frac{dy}{dx} = \frac{4\pi^2\left(8.99\times10^{-3}\right)}{3\left(2.109\times10^{-13}\right)} = 5.61 \times 10^{11}\ Pa$$

$$\frac{\Delta E_0}{E_0} = \frac{\Delta(dy/dx)}{dy/dx} + \frac{\Delta I}{I} = \frac{2.30\times10^{-4}}{8.99\times10^{-3}} + 5.18\% = 2.56\% + 5.18\% = 7.74\%$$

$$\Rightarrow\ E_0 = (5.61 \pm 0.43) \times 10^{11}\ Pa$$

Sample Calculation 10. Young's modulus at room temperature from the gradient of Figure 4 (Section 5.1).

Since no straight line passes through all error bars of Figure 4, the gradient uncertainty is taken as the standard error of the linear fit.

Since the infrared thermometer reading fluctuated during measurement (Section 6.1), its ±0.1 °C resolution underestimates the temperature uncertainty; the size of the fluctuation was not recorded, so this uncertainty could not be quantified and is treated

as a limitation. The gradient's uncertainty comes from the steepest and shallowest lines through the error bars in Figure 9, and is carried into α_E with Equation 13.

$$\frac{\Delta\alpha_E}{|\alpha_E|} = \frac{\Delta(df/dT)}{|df/dT|} + \frac{\Delta f_0}{f_0}$$

Equation 13. Percentage uncertainty in α_E, applying Equation 10 to α_E = (2/f₀)(df/dT) (Equation 14).

$$\Delta\left(\frac{df}{dT}\right) = \frac{(-0.00312)-(-0.00554)}{2} = 0.00121\ Hz\ ^{\circ}C^{-1}\ (25.8\%), \Delta f_0 = \frac{26.616-26.381}{2} = 0.117\ Hz\ (0.44\%)$$

Sample Calculation 11. Gradient and room-temperature frequency from the steepest and shallowest lines in Figure 9, evaluated at 24.8 °C.

$$\frac{\Delta\alpha_E}{|\alpha_E|} = 25.8\% + 0.44\% = 26.2\% \quad \Rightarrow \quad \alpha_E = (-3.54 \pm 0.93) \times 10^{-4}\ ^{\circ}C^{-1}$$

Sample Calculation 12. Temperature coefficient of Young's modulus (Section 7.2).

Quantity	Value	Absolute uncertainty	Percentage uncertainty	Sample Calculation
Tip mass, m_tip	1.225 g	±0.0025 g	0.20%	1
Second moment of area, I	2.109 × 10⁻¹³ m⁴	±0.109 × 10⁻¹³ m⁴	5.18%	2
Total face area, A_total	8548.0 mm²	±166.6 mm²	1.95%	3
Density, ρ	12,650 kg m⁻³	±390 kg m⁻³	3.06%	4
Beam term of Equation 4, ρAL	22.69 g	±1.45 g	6.40%	5
Effective mass, m_eff	6.574 g	±0.345 g	5.25%	6
Frequency of one trial, f (first trial at 82.1 °C)	26.2934 Hz	±0.057 Hz	0.22%	7
Mean frequency, f̄ (82.1 °C)	26.2567 Hz	±0.0696 Hz	0.27%	8
y = f² (L = 10 cm)	1905.6 Hz²	±18.1 Hz²	0.95%	9

Quantity	Value	Absolute uncertainty	Percentage uncertainty	Sample Calculation
Young's modulus at room temperature, E_0 (from Figure 4)	561 GPa	±43 GPa	7.7%	10
Gradient, df/dT	-0.00469 Hz °C⁻¹	±0.00121 Hz °C⁻¹	25.8%	11
Room-temperature frequency, f_0	26.52 Hz	±0.12 Hz	0.44%	11
Temperature coefficient of Young's modulus, α_E	-3.54×10^{-4} °C⁻¹	$\pm 0.93 \times 10^{-4}$ °C⁻¹	26.2%	12

Table 4. Uncertainties of the quantities derived by propagation, with the sample calculation for each.

Table 4 shows that the uncertainty in α_E comes almost entirely from the gradient (25.8 of 26.2%), and that the dimensional uncertainties (l, ρ, m_eff) do not enter it at all, as expected from Equation R4. They matter only for E_0 and the absolute frequency (Section 7.3).

6.3 Quantitative Data

The FFT in the data-logging software placed frequencies in bins about 0.49 Hz wide, larger than the whole predicted shift of about 0.6 Hz, so the trend could not be resolved. I therefore used Claude (Anthropic), an AI model ("Create a Guideline"), to write a Python script that measures each trial's frequency with sub-bin precision; Claude also proposed the method, which follows standard spectral analysis (Smith). I collected the data, ran the script and interpreted its results.

The raw magnetic field signal was exported from each .gambl file and processed in the following steps.

- **Separating flicks.** The signal's root-mean-square size was computed over 100 ms windows. Stretches more than three times the background noise level and at least 500 samples (0.5 s) long were treated as one flick, so that each flick became one trial.
- **Removing the offset.** The mean of each segment was subtracted, since the sensor reads the magnet's steady field as well as its oscillation.
- **Hann window.** Each segment was multiplied by a Hann window, which tapers it smoothly to zero at both ends. Without it, the abrupt start and end of the segment would spread the peak into neighbouring bins (spectral leakage).
- **Zero-padding.** Each segment was extended to eight times its length with zeros. This adds no new information but samples the spectrum eight times more finely, so the peak's shape is traced more clearly.
- **FFT and peak search.** The FFT was computed and the bin of largest magnitude found. For the temperature runs, segments whose peak lay outside 20 to 35 Hz were rejected; these peaked at 3 to 4.4 Hz, the tray's motion (Section 7.5).
- **Parabolic interpolation.** A parabola was fitted through the logarithms of the peak bin and its two neighbours, and its vertex taken as the frequency (Equation R5).

$$f = f_k + p\,\Delta f_{bin}, \qquad p = \frac{1}{2} \cdot \frac{lnM_{k-1} - lnM_{k+1}}{lnM_{k-1} - 2lnM_k + lnM_{k+1}}$$

Equation R5. Parabolic interpolation of the spectral peak (Smith). f_k is the frequency of the largest bin, M_{k-1}, M_k and M_{k+1} are the magnitudes of that bin and its

neighbours, Δf_bin is the zero-padded bin width and p, between -½ and ½, is the fraction of a bin by which the true peak is offset.

The method was checked on synthetic signals of known frequency (Appendix F). The full procedure is in Appendix E, and the results are in Table 5.

T (°C)	Mean frequency f̄ (Hz)	Combined uncertainty δf (Hz)	Standard deviation σ (Hz)	Accepted trials n
24.8	25.5620	±0.0450	0.0621	2
41.2	26.0559	±0.0612	0.0296	7
62.0	26.1264	±0.0696	0.0495	11
82.1	26.2567	±0.0696	0.0413	8
97.9	26.1358	±0.0751	0.0714	8
121.7	26.1348	±0.0558	0.0563	5
138.4	25.9461	±0.0808	0.1267	4

Table 5. Mean natural frequency at each temperature (L = 15 cm). Per-trial values are in Appendix E (Table E1).

7. Data Analysis

7.1 Temperature Trend

Fitting all seven points gives a positive gradient, opposite to the prediction, with R^2 = 0.19 (Table 6, Figure 8). The main cause is the 24.8 °C point, 0.4 to 0.7 Hz below every heated measurement. Since heating can only lower Young's modulus, the bench point cannot be explained by temperature: it must reflect a different clamping state, as it was measured on the bench, outside the oven, before the assembly was placed in the oven.

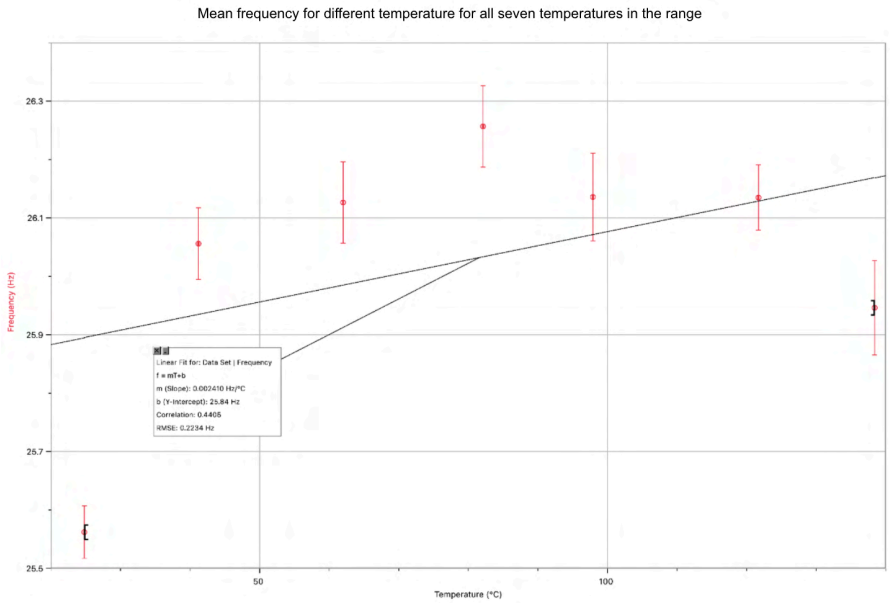


Figure 8. All seven points with error bars and a linear fit.

Data included	Points	Gradient df/dT (Hz °C^{-1})	R^2
All temperatures, 24.8 to 138.4 °C	7	+0.0024	0.19
Oven only, 41.2 to 138.4 °C	6	-0.00089	0.10
Oven, 82.1 to 138.4 °C	4	-0.00469 ± 0.00121	0.83

Table 6. Linear fits of frequency against temperature for three subsets of the data.

The 41.2 °C and 62.0 °C points also lie below the hotter points, so the frequency rises between 41.2 °C and 82.1 °C. A temperature-measurement error cannot explain this, as the 41.2 °C and 62.0 °C points would need to be about 80 °C and 46 °C hotter than recorded to lie on the trend of the hotter points. Since the Young's modulus of steel decreases with temperature, the increasing trend suggests that another variable, such as the clamping or support condition, changed during these trials; however, since the cause could not be identified, these points cannot be disregarded as errors. Therefore, no clear relationship between temperature and frequency is supported over the full

range. The four points from 82.1 °C to 138.4 °C do decrease steadily ($R^2 = 0.83$), consistent with the prediction, so they are analysed separately in Section 7.2 to estimate the temperature coefficient for this range only.

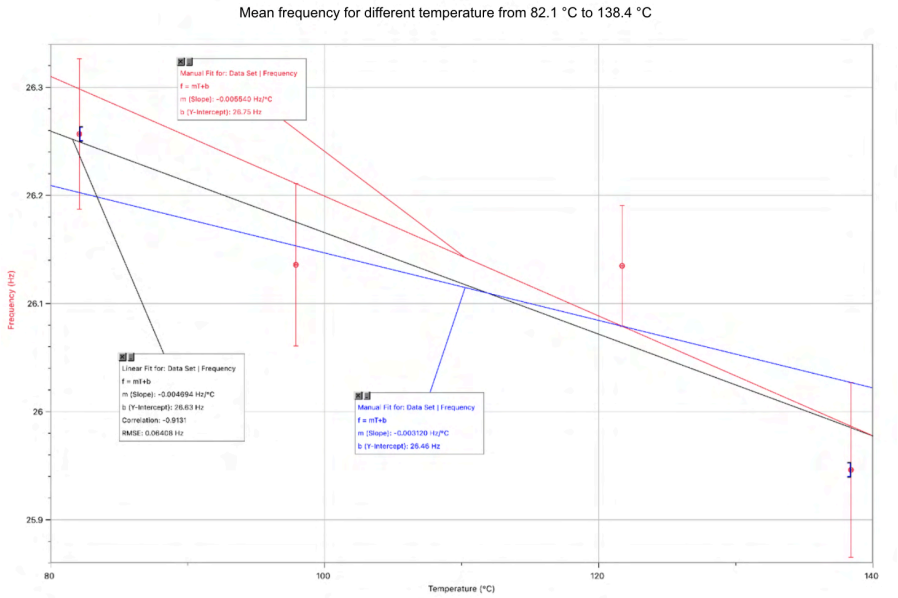


Figure 9. The four points from 82.1 °C with error bars, the best-fit line, and the steepest and shallowest lines through all four error bars.

7.2 Temperature Coefficient

The best-fit gradient of the four points is -0.00469 Hz $°C^{-1}$. The steepest line through all four error bars has gradient -0.00554 Hz $°C^{-1}$ and intercept 26.7532 Hz, and the shallowest has -0.00312 Hz $°C^{-1}$ and 26.4587 Hz, giving an uncertainty of ±0.00121 Hz $°C^{-1}$ (Sample Calculation 11).

To turn the gradient into α_E, Equation 8 is written in terms of f_0, its value at T_0 (Section 3.3). The two constant factors in front of the bracket are exactly f_0, so Equation 8

becomes f ≈ f₀(1 + ½α_E(T − T₀)). This is a straight line in T, and differentiating gives its gradient.

$$f \approx f_0\left(1 + \frac{1}{2}\alpha_E T - \frac{1}{2}\alpha_E T_0\right) \Rightarrow \frac{df}{dT} = \frac{1}{2}\alpha_E f_0$$

Equation R6. Gradient of Equation 8, with f₀ as defined in Section 3.3.

Taking T₀ = 24.8 °C, the room temperature of this experiment, the best-fit line gives f₀ = 26.6349 − 0.004694 × 24.8 = 26.52 Hz, with ±0.12 Hz from the steepest and shallowest lines (Sample Calculation 11). Rearranging Equation R6 gives Equation 14.

$$\frac{df}{dT} = \frac{1}{2}\alpha_E f_0 \quad \Rightarrow \quad \alpha_E = \frac{2}{f_0} \cdot \frac{df}{dT} = \frac{2\times\left(-0.00469\ Hz\ ^\circ C^{-1}\right)}{26.52\ Hz} = -\ 3.54\ \times 10^{-4}\ ^\circ C^{-1}$$

Equation 14. Temperature coefficient of Young's modulus from the gradient of Equation 8 (Section 3.3); its uncertainty is found in Sample Calculation 12.

So the estimated α_E = (-3.5 ± 0.9) × 10⁻⁴ °C⁻¹ for this range. The f₀ used here is taken from the measurements, not from the model, so α_E does not depend on E₀ or on any dimension of the ruler (Equation R4).

Source This investigation, 82.1 to 138.4 °C	α_E (°C⁻¹) (-3.54 ± 0.93) × 10⁻⁴
Literature value for steel (Latella and Humphries)	about -4.0 × 10⁻⁴

Table 7. Experimental and literature α_E. The difference is 11.5%, and the literature value lies within the experimental range of -2.61 to -4.47 × 10⁻⁴ °C⁻¹.

The percentage difference, |(-3.54) − (-4.0)| / 4.0 = 11.5%, is smaller than the 26.2% uncertainty, so the estimate is consistent with the literature value. However, this result should be treated with caution, since it is based on only four points in a restricted range.

7.3 Comparison with the Model

The model can be tested in two ways: the fractional gradient, which depends only on α_E, and the absolute frequency, which depends on every constant.

The measured fractional gradient is (-0.00469)/26.52 = -1.77 × 10^{-4} $°C^{-1}$, and the model with the literature α_E predicts ½α_E = -2.0 × 10^{-4} $°C^{-1}$ (Equation R4). These agree within the 26% uncertainty, so within this restricted range the temperature dependence of the model is consistent with the data.

The absolute level does not agree: using E_0 from Section 5.1 in Equation 4, the model predicts f_0 = 20.1 Hz instead of the measured 26.5 Hz, 24% too low. The Rayleigh approximation accounts for only 1% of this (Equation R3), and in the wrong direction, since it overestimates f. The offset must therefore come from the constants, E_0, the dimensions or the clamp, rather than from the temperature dependence.

Clamp compliance. If the textbook clamp of Section 5.1 were rigid, Equation 4 predicts that frequency should scale with length as f ∝ L^-1.915 over 10 to 30 cm, steeper than $L^{-1.5}$ because the beam's own mass grows with L. The length data instead follow f ∝ L^-1.767. A shallower exponent is what a slightly compliant clamp produces: a ruler that can rotate a little where it leaves the textbooks bends as if it were longer by a fixed amount δ, and a fixed δ is a larger fraction of a short beam, lowering its frequency the most. This is also why the short lengths fall below the line in Figure 4. Allowing for such an offset gives Equation 15.

$$f = \frac{C}{\sqrt{(L+\delta)^3\left[\frac{33}{140}\rho A\,(L+\delta)+m_{tip}\right]}}, \qquad C = \frac{1}{2\pi}\sqrt{3E_0 I}$$

Equation 15. Equation 4 with a constant effective length offset δ for the textbook clamp; C and δ were fitted to the five length measurements.

Fitting C and δ to the five length measurements gives δ = 15 ± 3 mm, where the range is the set of δ values that keep the RMS residual below 2%. The corrected E_0 then predicts 26.62 Hz at 15 cm, within 0.4% of the oven series (Table 8).

Quantity	Without offset (δ = 0)	With offset (δ = 15 mm)
RMS residual of the length fit	9.5%	0.9%
E_0 from the length data (h = 0.46 mm)	5.61×10^{11} Pa (Section 5.1)	9.81×10^{11} Pa
Model f_0 at L = 15.00 cm	20.1 Hz	26.62 Hz
Difference from the oven-series f_0 of 26.52 Hz	-24%	+0.4%

Table 8. Effect of a clamp offset on the length fit and the predicted room-temperature frequency. δ = 15 ± 3 mm keeps the RMS residual below 2%.

The G-clamped wooden blocks used in the oven were much stiffer than the textbooks, which is why the oven series sits close to the corrected model at the nominal 15 cm.

Thickness. Even with the offset, E_0 = 981 GPa is about five times the accepted value, and the density of 12,650 kg m^{-3} is far above the accepted 7,500 to 8,000 kg m^{-3} (Appendix D; “Young’s Modulus”). Both depend on the measured thickness (Equation 16): the density is the mass divided by A_total · h, while E_0 comes from $I = bh^3/12$.

$$\rho \propto h^{-1}, \qquad E_0 \propto h^{-3}$$

Equation 16. Dependence of the calculated density and E_0 on the measured thickness, since ρA is independent of h (Sample Calculation 5) while $I \propto h^3$.

An underestimated thickness therefore inflates both, E_0 much more strongly. Choosing the thickness that makes the density 7,900 kg m⁻³ gives h = 0.46 × 12,650/7,900 = 0.737 mm, and this single value brings E_0 to between 136 and 239 GPa, depending on whether the clamp offset is included, which brackets the accepted 193 to 200 GPa (Table 9). One correction fixing two independent quantities suggests the micrometer readings underestimate the mean thickness, for example if they were taken where the ruler is thinnest, or on its engraved scale.

Thickness used (mm)	Density (kg m⁻³)	E_0 from length data, no offset (GPa)	E_0 from length data with offset (GPa)
0.46 (measured)	12,650	561 ± 43	981
0.53 (remeasured)	10,980	367	642
0.737 (inferred from a density of 7,900 kg m⁻³)	7,900	136	239

Table 9. Density and E_0 for three thicknesses. Accepted values for stainless steel are 7,500 to 8,000 kg m⁻³ and 193 to 200 GPa.

The thickness does not affect α_E, since it cancels in the fractional gradient (Equation R4). The clamp, however, could: if the clamping condition changed with temperature inside the oven, as the low-temperature points suggest it may have, it would also change the gradient and therefore the estimated α_E.

Thermal expansion. Uniform expansion changes f by only +0.1% over the range (Equation R1, Section 2.3).

$$f \propto \frac{1}{L^2}\sqrt{\frac{EI}{\rho A}}, \quad L, b, h \propto s,\ \rho A \propto s^{-1},\ I \propto s^4 \quad \Rightarrow \quad f \propto s^{1/2} \approx 1 + \frac{1}{2}\alpha_{th}\,\Delta T$$

Equation 17. Effect of uniform thermal expansion on Equation 3, where every length scales by s = 1 + α_th ΔT, α_th is the linear expansion coefficient and the mass is

unchanged. With $\alpha_{th} \approx 1.7 \times 10^{-5}$ °C⁻¹ for stainless steel, the rise over 113.6 °C is +0.10%, compared with -2.3% from the change in E.

Over the 56.3 °C range actually used (82.1 to 138.4 °C), expansion would raise f by 0.05%, or 0.013 Hz, against a measured fall of 0.26 Hz. Ignoring it makes the measured $|\alpha_E|$ about 5% too small, a bias well within the 26% uncertainty.

7.4 Scatter Between Trials

The frequencies of individual trials at the same temperature vary considerably: at 138.4 °C they range from 25.86 Hz to 26.13 Hz, a spread of 0.27 Hz (Appendix E), comparable to the entire decrease of 0.31 Hz between 82.1 °C and 138.4 °C and equivalent to about 58 °C of heating at the measured gradient. Since temperature, length and clamping were constant within each recording, the most likely cause is the initial displacement, which varied because the ruler was flicked by hand (Section 6.1). A cantilever's frequency can depend slightly on amplitude, particularly when the clamp contact is not perfectly rigid, so a stronger flick may give a slightly different frequency. This is therefore a significant source of random uncertainty, and averaging several trials at each temperature reduces but does not remove it.

7.5 Coupling to the Oven Tray

The rejected segments peaked at 3.0 to 4.4 Hz (Appendix E), the visible motion of the tray in Section 6.1. A peak at 3.0 Hz also appears in one 10 cm bench recording, so it belongs to whatever surface supports the clamp rather than to the oven alone. Because this frequency is about eight times below the ruler's 26 Hz, the two motions are far from resonance and exchange little energy, and the accepted segments show a single clean

peak in the 20 to 35 Hz window. The tray's motion therefore does not shift the measured frequency directly. However, a support that moves means the clamp is not a perfectly fixed end, which adds to the compliance discussed in Section 7.3; this would affect the absolute frequency more than the gradient, provided the coupling did not change with temperature.

8. Conclusion

Over the full range from 24.8 °C to 138.4 °C, no clear relationship between temperature and the natural frequency of the 15 cm stainless steel cantilever was found: a linear fit of all seven points gives a positive gradient with $R^2 = 0.19$, and the frequency rose between 41.2 °C and 82.1 °C, which the model cannot explain. Only the four points from 82.1 °C to 138.4 °C show the decrease predicted by the model, at -0.0047 ± 0.0012 Hz $°C^{-1}$. This is consistent with the mathematical model, in which heating lowers Young's modulus and the frequency falls with its square root (Section 2.3, Equation R4). From these four points, the temperature coefficient of Young's modulus is estimated as (-3.5 ± 0.9) × 10^{-4} $°C^{-1}$, similar to the literature value of about -4.0 × 10^{-4} $°C^{-1}$ (Latella and Humphries). This is an estimate for a limited range rather than a confirmation of the model, since the cause of the behaviour below 82.1 °C was not identified and may also have affected the higher temperatures.

The model also did not predict the absolute frequency, giving 20.1 Hz instead of the measured 26.5 Hz. The calculated density and Young's modulus differ greatly from the accepted values for stainless steel, which indicates errors in the measured constants,

most likely the thickness, and a compliant clamp in the length experiment (Section 7.3); comparison of the model's absolute values with the data is therefore strongly limited.

The research question is therefore answered only partially: between 82.1 °C and 138.4 °C, the natural frequency decreased with temperature in the way the model predicts, but a clear relationship over the whole range was not established. Since seasonal temperature changes can shift the natural frequencies of steel structures by amounts comparable to those caused by damage (Luo et al.; Xia et al.), confirming this relationship with a more controlled experiment remains worthwhile.

9. Evaluation

9.1 Strengths

The frequency-extraction method was a major strength. The software's FFT could not resolve shifts smaller than about 0.49 Hz, but windowing, zero-padding and interpolation resolved each flick to a few hundredths of a hertz, and the method was checked on synthetic signals of known frequency (Appendix F). Without it, the 0.26 Hz effect could not have been measured at all.

The analysis was designed so that the main result does not depend on the least certain quantities. Because α_E comes from the fractional gradient (Equation R4), the large errors in E_0, the density and the thickness, which were discovered during the analysis, do not affect it, and almost all of its 26% uncertainty comes from the gradient itself.

Finally, unexpected results were investigated rather than ignored: a temperature-measurement error was tested and ruled out as the cause of the low

points, and the discrepancy in absolute frequency was traced to likely causes in the clamp and the thickness measurement, each of which made a prediction (the length exponent, the density) that the data supported.

9.2 Limitations and Systematic Errors

Limitation	Effect on the results	Improvement
Unexplained low-temperature points and change of clamp	No clear relationship over the full range; the cause of the low 41.2 °C and 62.0 °C values is unknown and may also affect the hotter points; E_0 from the textbook clamp could not be transferred to the oven clamp (Sections 7.1, 7.3)	Use one rigid clamp throughout; take the room-temperature reading in the unheated oven; repeat in smaller steps during both heating and cooling
Initial displacement by hand	Spread between trials of up to 0.27 Hz, comparable to the temperature effect (Section 7.4)	Release the tip from a fixed stop
Infrared thermometer	Inaccurate below 50 °C and fluctuating, so the temperature uncertainty could not be quantified; opening the door cooled the ruler (Section 6.1)	Thermocouple taped to the ruler, read with the door closed
Only four points in the analysed range	Gradient uncertainty of 25.8% dominates α_E	Smaller temperature steps and repeated runs
Tray vibration	Rejected trials and a slightly moving support, without shifting the measured frequency (Section 7.5)	Suspend or isolate the assembly from the tray
Thickness measurement	Density and E_0 far too high; no effect on α_E (Section 7.3)	Measure the thickness at many points and check the result against the density
Rayleigh approximation and neglected thermal expansion	About +1% on the absolute frequency and about -5% on the magnitude of α_E; both well within the uncertainty	Use the exact tip-mass equation (Equation R2) and include Equation R1 in the model

Table R2. Limitations in order of their effect on the result, with improvements.

The main limitation is that no clear relationship was observed over the full temperature range, and the agreement with the model depends on only four data points. Since the

cause of the low 41.2 °C and 62.0 °C values was not identified, it may also have affected the analysed points, so the estimated temperature coefficient applies only from 82.1 °C to 138.4 °C. Repeating the measurements in smaller temperature steps, during both heating and cooling, would show whether this behaviour is reproducible.

The largest procedural error is the uncontrolled initial displacement of the ruler. Although the frequency varies by only about 1% between flicks, the whole temperature effect is of similar size, so this remains a significant source of scatter even after averaging; releasing the tip from a fixed stop would remove it.

The FLIR camera could not be used inside the oven, the infrared thermometer reading fluctuated, and opening the door cooled the ruler. A thermocouple taped to the ruler and read with the door closed would improve accuracy and reliability, and performing the room-temperature trial in the unheated oven with the same clamp would remove the difference between setups. Finally, the Rayleigh approximation overestimates the frequency by about 1%, and the thickness should be measured at many points and checked against the resulting density; neither affects α_E.

9.3 Extensions

The possible amplitude dependence suggested in Section 7.4 could be investigated directly by releasing the ruler from a series of controlled displacements at a fixed temperature, which would show whether it comes from the clamp or the beam. A furnace with a thermocouple would extend the range beyond 140 °C and show where the linear model of E(T) begins to fail. Other metals, such as aluminium or brass, would test whether the method gives their known α_E values, and the second vibration mode,

at about 6.3 times the frequency ($\beta_2 L = 4.6941$ in Appendix B), would test whether the same α_E is obtained from a different mode shape, since Equation R4 predicts the same fractional gradient for every mode.

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Appendix A. Derivation of Second Moment of Area for Rectangular Cantilever Beam

As mentioned, the second moment of area is a second-dimension geometrical property that shows how the object's points are dispersed in an arbitrary axis in the cross-sectional plane. The following is a view of the cross sectional area with a cantilever beam width 'b' and height 'h', same with the cantilever shown in Figure 2.

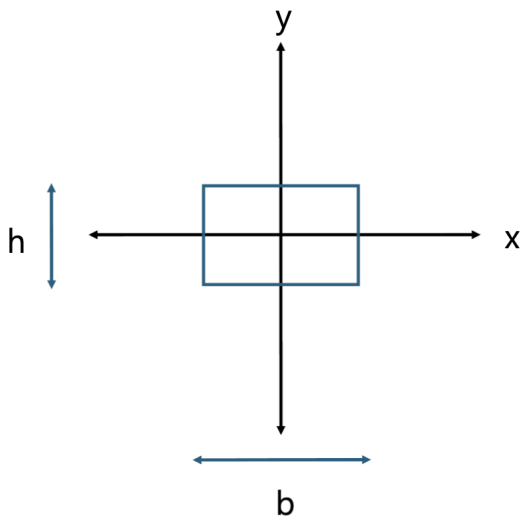


Figure A1. Diagram of cross-sectional area of a cantilever beam from Figure 2.

The equation of second moment of area I [m⁴] is given as

$$I = \int y^2 \, dA$$

where y is the distance from the neutral axis (x-axis for the diagram) and dA is an infinitesimal area for the area given dA = dx · dy.

For the diagram above, it can be derived by

$$I = \int y^2 \, dA = \int_{-h/2}^{h/2} y^2 \cdot b \cdot dy = b\left[\frac{y^3}{3}\right]_{-h/2}^{h/2} = \frac{bh^3}{12}$$

Equation A1. Second moment of area of a rectangular cross-section.

Appendix B. Natural Frequency of First Mode Cantilever Vibration

Derivation from Euler-Bernoulli Beam Theory

The Euler-Bernoulli Beam Theory says

$$EI\frac{\partial^4 w}{\partial x^4} + \rho A\frac{\partial^2 w}{\partial t^2} = 0$$

with w(x, t) being the transverse displacement from the neutral axis of vibration. To acquire the general solution of w(x, t), the function of form w(x, t) = W(x) · T(t) (where W(x) and T(t) are independent functions of displacement from the fixed end 'x' and time 't') should be assumed. This assumption is valid since the shape of the ruler does not change during vibration, meaning that the same vibrating motion is shown in every displacement from the free end (as a function of time) but only the amplitude differs (as a function of x).

Assuming this form of solution gives

$$\frac{\partial^4 w}{\partial x^4} = W''''(x) \cdot T(t), \qquad \frac{\partial^2 w}{\partial t^2} = W(x) \cdot T''(t)$$

These terms can be substituted into the original governing equation and manipulated.

$$EI \cdot W''''(x) \cdot T(t) + \rho A \cdot W(x) \cdot T''(t) = 0\,; \; \frac{EI \cdot W''''(x) \cdot T(t) + \rho A \cdot W(x) \cdot T''(t)}{W(x) \cdot T(t)} = 0$$

$$EI \cdot \frac{W''''(x)}{W(x)} + \rho A \cdot \frac{T''(t)}{T(t)} = 0; \; \frac{EI}{\rho A} \cdot \frac{W''''(x)}{W(x)} = -\frac{T''(t)}{T(t)} = const$$

The two terms are equal to a constant for a function of x and a function of t to equal each other at every possible x and t. But this means that

$$-\frac{T''(t)}{T(t)} = const \quad \Rightarrow \quad T''(t) = -const \cdot T(t)$$

This equation is in the form of an equation for SHM. The acceleration equivalent term of T''(t) is directly proportional to the displacement equivalent term of T(t). Thus, it could be concluded that the constant term is angular frequency squared (ω^2), which is shown with the IB Physics equation $a = -\omega^2 x$. So,

$$\frac{EI}{\rho A} \cdot \frac{W''''(x)}{W(x)} = \omega^2 \quad \Rightarrow \quad W''''(x) = \frac{\rho A \omega^2}{EI} W(x)$$

and setting $\beta^4 = \rho A\omega^2/EI$ for simplicity,

$$W''''(x) - \beta^4 W(x) = 0$$

From here, the general solution W(x) = e^(λx) is assumed to determine the general solution.

$$\frac{d^4}{dx^4} e^{\lambda x} - \beta^4 e^{\lambda x} = 0;\ \lambda^4 e^{\lambda x} - \beta^4 e^{\lambda x} = 0, \quad \left(e^{\lambda x} \neq 0\right) \lambda^4 - \beta^4 = 0;\ \lambda = \pm\beta, \ \pm i\beta$$

This gives the general solution using the principle of superposition

$$W(x) = C_1 e^{\beta x} + C_2 e^{-\beta x} + C_3 e^{i\beta x} + C_4 e^{-i\beta x}$$

which can be rewritten using the definition of sinusoidal and hyperbolic functions regarding Euler's identity

$$W(x) = C_1 cosh(\beta x) + C_2 sinh(\beta x) + C_3 cos(\beta x) + C_4 sin(\beta x)$$

and its derivatives of

$$W'(x) = \beta\left[C_1 sinh(\beta x) + C_2 cosh(\beta x) - C_3 sin(\beta x) + C_4 cos(\beta x)\right] W''(x)$$

$$= \beta^2\left[C_1 cosh(\beta x) + C_2 sinh(\beta x) - C_3 cos(\beta x) - C_4 sin(\beta x)\right] W'''(x)$$

$$= \beta^3\left[C_1 sinh(\beta x) + C_2 cosh(\beta x) + C_3 sin(\beta x) - C_4 cos(\beta x)\right]$$

To acquire a particular solution of this differential equation, a few conditions should be physically explained:

- W(0) = 0, as there is no transverse displacement of the ruler at the fixed end since it is fixed with clamps with immense pressure.
- W'(0) = 0, as the ruler is completely horizontal and has no slope at the fixed end due to the immense pressure of the clamp holding the ruler flat.
- W''(L) = 0, as W''(x) is the bending moment (internal rotational tendency) due to mass after the position x, but there is no mass of ruler beyond the end of the ruler at x = L.
- W'''(L) = 0, as W'''(x) is the shear force (internal tendency to cut transversely) due to force acting beyond the position x, but there is no ruler or mass for force to act on beyond x = L.

Using the definition of sinusoidal and hyperbolic functions, the boundary conditions can be applied.

$$W(0) = C_1 + C_3 = 0 \quad \Rightarrow \quad C_3 = -C_1 \; W'(0) = C_2 + C_4 = 0$$

$$\Rightarrow \quad C_4 = -C_2 \; W''(L) = C_1 cosh(\beta L) + C_2 sinh(\beta L) + C_1 cos(\beta L) + C_2 sin(\beta L) = 0$$

$$\Rightarrow \; C_1(cosh(\beta L) + cos(\beta L)) + C_2(sinh(\beta L) + sin(\beta L)) = 0 \quad \ldots$$

$$(i) \; W'''(L) = C_1 sinh(\beta L) + C_2 cosh(\beta L) - C_1 sin(\beta L) + C_2 cos(\beta L) = 0$$

$$\Rightarrow\ C_1(sinh(\beta L) - sin(\beta L)) + C_2(cosh(\beta L) + cos(\beta L)) = 0 \quad \ldots\ (ii)$$

Substituting P = cosh(βL) + cos(βL), Q = sinh(βL) + sin(βL) and R = sinh(βL) − sin(βL) gives

$$C_1 P + C_2 Q = 0 \quad \ldots\ (i), \qquad C_1 R + C_2 P = 0 \quad \ldots\ (ii)$$

which through solving the system of equations is able to derive (with C_2 being non-zero since vibration occurs physically)

$$P^2 - QR = 0;\ (cosh(\beta L) + cos(\beta L))^2 - (sinh(\beta L) + sin(\beta L))(sinh(\beta L) - sin(\beta L)) = 0$$

$$cosh^2(\beta L) + 2cosh(\beta L)cos(\beta L) + cos^2(\beta L) - sinh^2(\beta L) + sin^2(\beta L) = 0\ ;\ 2 + 2cosh(\beta L)cos(\beta L) = 0$$

$$cos(\beta L)cosh(\beta L) + 1 = 0$$

Equation B1. Characteristic equation of the fixed-free cantilever.

This characteristic equation gives the values βL = 1.8751, 4.6941, 7.8548, 10.996, and so on. The different values represent each solution for different modes of vibration, increasing with the number of modes. But since my method of applying initial displacement mostly creates the first mode; higher modes dissipate energy faster with air damping, decaying quickly; and the frequency difference is large enough to experimentally differentiate with the FFT (fast Fourier transform), the first mode is used.

$$\beta_1 L \approx 1.8751, \qquad \beta = \frac{\beta_1 L}{L}$$

Recalling the definition of β,

$$\beta^4 = \frac{\rho A \omega^2}{EI} \quad \Rightarrow \quad \omega^2 = \frac{\beta^4 EI}{\rho A} = \frac{(\beta_1 L)^4}{L^4} \cdot \frac{EI}{\rho A}$$

Using the formula for frequency in IB Physics,

$$f_1 = \frac{\omega}{2\pi} = \frac{(\beta_1 L)^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}} = \frac{1.8751^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

Equation B2. Natural frequency of the fundamental mode (Equation 3 in the main text).

However, this derivation does not consider the point mass of the magnets attached at the end of the ruler, which introduces inaccuracy. Thus, the Rayleigh method is used to fix this error; however, the derivation is necessary to acquire the solution of angular frequency that is used in the Rayleigh method.

Appendix C. Rayleigh Method for Approximation

From IB Physics, the natural frequency of a system is given as

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m_{eff}}}$$

in which k_eff and m_eff are the effective stiffness and mass of a system, such as the representation of multiple springs in a mass-spring system as a single spring constant. The same applies for this system too.

Since the beam's mass is spread continuously along its length, an approximate equivalent point-mass should be derived to evaluate the system with the addition of the point-masses at the tip of the ruler. This can be derived by equating the kinetic energy of the entire beam to that of an equivalent point mass.

$$dKE = \frac{1}{2}(\rho A \cdot dx)\left(\frac{\partial w(x,t)}{\partial t}\right)^2$$

where dKE is the kinetic energy of an infinitesimal element of the beam with length dx. The term ρA · dx represents the mass of that element, as ρ = m/V and A is the

cross-sectional area, so ρA is the mass per unit length. The term ∂w(x,t)/∂t is the velocity, the first derivative of displacement with respect to time. As stated earlier, w(x, t) = W(x)T(t), where T(t) is a sinusoidal SHM function; taking T(t) to have unit amplitude, the maximum of its time derivative is ω, so the maximum velocity of each element is ω · W(x). So,

$$dKE = \frac{1}{2}\rho A \cdot dx \cdot \omega^2 \cdot W(x)^2$$

and integrating both sides,

$$KE = \frac{1}{2}\rho A \omega^2 \int_0^L W(x)^2 \, dx$$

which is the maximum kinetic energy of the beam. The effective mass m_eff of the beam as a point-mass is found by giving it the maximum velocity of the end of the beam, v_max = ωW(L), since W(x) is largest at the free end, and equating its kinetic energy with that of the beam.

$$\frac{1}{2} m_{eff} \omega^2 W(L)^2 = \frac{1}{2}\rho A \omega^2 \int_0^L W(x)^2 \, dx \quad \Rightarrow \quad m_{eff} = \rho A \int_0^L \left(\frac{W(x)}{W(L)}\right)^2 dx$$

Equation C1. Effective mass of the beam for an assumed vibration shape W(x).

Using the exact mode shape. From Appendix B,

$$\frac{W(x)}{W(L)} = \frac{C_1(cosh(\beta x) - cos(\beta x)) + C_2(sinh(\beta x) - sin(\beta x))}{C_1(cosh(\beta L) - cos(\beta L)) + C_2(sinh(\beta L) - sin(\beta L))}, \qquad \frac{C_2}{C_1} = -\frac{cosh(\beta L) + cos(\beta L)}{sinh(\beta L) + sin(\beta L)} \approx -0.7341$$

with βL ≈ 1.8751, from Appendix B equation (i). Evaluating Equation C1 with this shape gives exactly m_eff = 0.25 m_beam. However, this mode shape belongs to the beam without a tip mass, and pairing it with the static stiffness derived below would mix two

different shapes. Rayleigh's method instead uses one assumed shape, the static deflection curve, for both the stiffness and the kinetic energy.

Static deflection shape and stiffness. Using the static version of the Euler-Bernoulli Beam Theory,

$$EI\frac{d^4w}{dx^4} = 0 \quad \Rightarrow \quad EI\,w(x) = \frac{C_1x^3}{6} + \frac{C_2x^2}{2} + C_3x + C_4$$

By considering a force F exerted downwards on the beam at the end, x = L, and using the same conditions as before due to the properties of a cantilever, w(0) = 0, w'(0) = 0, w''(L) = 0 and EI · w'''(L) = −F, the particular solution is

$$w(x) = \frac{F}{6EI}\left(3Lx^2 - x^3\right), \qquad w(L) = \frac{FL^3}{3EI} \quad \Rightarrow \quad k_{eff} = \frac{F}{\Delta x} = \frac{3EI}{L^3}$$

Effective mass from the static shape. Normalising the static deflection gives φ(x) = w(x)/w(L) = (3Lx² − x³)/(2L³), with φ(0) = 0 and φ(L) = 1. Substituting into Equation C1:

$$m_{eff} = \rho A\int_0^L\left(\frac{3Lx^2-x^3}{2L^3}\right)^2 dx = \frac{\rho A}{4L^6}\int_0^L\left(9L^2x^4 - 6Lx^5 + x^6\right)dx = \frac{\rho A}{4L^6}\left(\frac{9L^7}{5} - L^7 + \frac{L^7}{7}\right)$$

$$= \frac{\rho A}{4L^6}\cdot\frac{33L^7}{35} = \frac{33}{140}\rho AL \approx 0.2357\,m_{beam}$$

Equation C2. Effective mass of the beam from the static deflection shape.

Thus, the total equivalent mass including the tip is (33/140)ρAL + m_tip, and

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{eff}}{m_{eff}}} = \frac{1}{2\pi}\sqrt{\frac{3EI/L^3}{\frac{33}{140}\rho AL + m_{tip}}}$$

Equation C3. Natural frequency with a tip mass by the Rayleigh method (Equation 4 in the main text).

This approximation can be verified by setting m_tip = 0, which should approximately equal the Euler-Bernoulli Beam Theory formula for frequency.

$$f_{Rayleigh} = \frac{1}{2\pi L^2}\sqrt{\frac{3}{33/140}}\sqrt{\frac{EI}{\rho A}} \approx 3.568 \cdot \frac{1}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}, \qquad f_{EBBT} = \frac{1.8751^2}{2\pi L^2}\sqrt{\frac{EI}{\rho A}} \approx 3.516 \cdot \frac{1}{2\pi L^2}\sqrt{\frac{EI}{\rho A}}$$

The two agree to within 1.5%, with the Rayleigh value slightly higher, as expected because Rayleigh's method gives an upper estimate of the frequency . Using the exact-mode value of 0.25 with the static stiffness instead would give 3.464, 1.5% too low.

Appendix D. Density of Ruler

The density of a three-dimensional object is defined as mass per unit volume in kg m^{-3}. As the ruler has a completely flat surface, the volume is equal to the surface area multiplied by the height.



Figure D1. Picture of the curved part of the ruler.

But due to the irregular shape at the end of the ruler, methods have been used to calculate the curved part. The total surface area of the ruler can be written as

$$A_{total} = A_{rectangle} + A_{curvature} - A_{circle}$$

By measuring with a ruler and using software measuring pixels, the length of the rectangular part of the ruler (l), the width of the ruler (w), the diameter of the circle (2r), and the height of the curvature (h_c) were measured. The respective areas were calculated in this way, with the curvature assumed to be a chord of an extended circle:

$$A_{rectangle} = w \cdot l = 26\,mm \cdot (300 + 27)\,mm = 8502\,mm^2$$

$$A_{circle} = \pi r^2 = \pi\left(\frac{6.25\,mm}{2}\right)^2 = 30.68\,mm^2 \; R = \frac{w^2}{8h_c} + \frac{h_c}{2} = 21.68\,mm$$

$$(radius\ of\ the\ extended\ circle\ for\ the\ area\ of\ the\ chord)$$

$$A_{curvature} = R^2 arccos\left(\frac{R-h_c}{R}\right) - \left(R - h_c\right)\sqrt{2Rh_c - h_c^2} = 76.69\,mm^2$$

$$A_{total} = 8548.01\,mm^2 \; V = A \cdot h = 8548.01\,mm^2 \cdot 0.46\,mm = 3932.0846 \times 10^{-9}\,m^3$$

$$\rho = \frac{m}{V} = 12650\,kg\,m^{-3}$$

Equation D1. Derivation of the density of the ruler.

However, a notable point is that the density of the ruler made out of stainless steel is too large. The accepted density of usual stainless steel is 7,500 to 8,000 kg m^{-3}, and the experimentally calculated density is far from it. This might suggest a procedural error that was present when measuring the dimensions of the ruler or the mass of it.

Appendix E. Frequency Extraction Method and Raw Data

Method

The Python script written with Claude (Section 6.3) processes each 20-second recording of the magnetic field in six steps.

- **Trial segmentation.** A trial was identified wherever the rolling root-mean-square amplitude of the signal, over a 100 ms window, exceeded three times the noise level of the quietest 20% of the recording. Segments shorter than 0.5 s were discarded.
- **Offset removal.** The mean of each segment was subtracted, removing the constant field of the tip magnets.
- **Windowing.** Each segment was multiplied by a Hann window (Equation E1), which tapers both ends to zero and reduces spectral leakage, the spreading of the peak into neighbouring bins because a segment does not hold a whole number of cycles.
- **Zero-padding and FFT.** Each segment was padded with zeros to eight times its length before the FFT. This adds no information but samples the spectrum eight times more finely, so a 1.5 s trial's bin spacing falls from 0.67 Hz to about 0.08 Hz.
- **Parabolic interpolation.** A parabola was fitted through the logarithms of the largest bin and its two neighbours, and its apex was taken as the frequency (Equations E2 and E3).
- **Rejection and averaging.** Trials whose peak lay outside the expected range (20 to 35 Hz for the temperature experiment) were rejected as tray vibration, and the accepted trials at each condition were averaged (Equation E4), with the combined uncertainty described in Section 6.2.

$$w_n = \frac{1}{2}\left(1 - cos\frac{2\pi n}{N-1}\right), \qquad n = 0, 1, \ldots, N-1$$

Equation E1. Hann window applied to a segment of N samples.

$$p = \frac{1}{2} \cdot \frac{lnM_1 - lnM_3}{lnM_1 - 2lnM_2 + lnM_3}$$

Equation E2. Fractional offset of the true peak from the largest bin, where M_2 is the magnitude of the largest bin and M_1 and M_3 are its neighbours (Smith).

$$f = f_{bin} + p\,\Delta f_{pad}$$

Equation E3. Interpolated peak frequency, where f_bin is the frequency of the largest bin and Δf_pad is the zero-padded bin spacing.

$$f^{-} = \frac{1}{n}\sum_{i=1}^{n} f_i$$

Equation E4. Mean frequency of the n accepted trials.

Raw data

In both tables, ± interpolation is the per-trial uncertainty ε = Δf_raw / 10 used in Section 6.2, and ± raw bin is Δf_raw / 2, the uncertainty without interpolation, shown for comparison. The 2 °C refrigerator trial is listed for completeness but excluded from the analysis (Section 6.1).

T (°C)	Set	Trial	f (Hz)	± interpolation (Hz)	± raw bin (Hz)	Duration (s)	Samples	Status
2	Set 1	1	3.0465	0.0762	0.3808	1.3	1313	Rejected
2	Set 1	2	3.0631	0.0943	0.4713	1.1	1061	Rejected
2	Set 1	3	26.2323	0.0163	0.0813	6.1	6147	Accepted
2	Set 1	4	25.9506	0.0765	0.3826	1.3	1307	Accepted
2	Set 1	5	26.0713	0.0664	0.332	1.5	1506	Accepted

T (°C)	Set	Trial	f (Hz)	± interpolation (Hz)	± raw bin (Hz)	Duration (s)	Samples	Status
2	Set 1	6	3.0675	0.0876	0.4382	1.1	1141	Rejected
2	Set 2	1	26.0769	0.0838	0.4191	1.2	1193	Accepted
2	Set 2	2	26.1395	0.0859	0.4296	1.2	1164	Accepted
2	Set 2	3	26.145	0.0298	0.1491	3.4	3354	Accepted
2	Set 2	4	26.1974	0.0943	0.4717	1.1	1060	Accepted
2	Set 2	5	26.2437	0.0838	0.4191	1.2	1193	Accepted
2	Set 2	6	26.1914	0.0829	0.4146	1.2	1206	Accepted
2	Set 2	7	26.35	0.1058	0.5291	0.9	945	Accepted
2	Set 2	8	26.2643	0.0909	0.4545	1.1	1100	Accepted
2 mean			**26.1693**	**±0.0812** (combined)			n = 11	
24.8	Set 1	1	25.6059	0.014	0.0701	7.1	7131	Accepted
24.8	Set 2	1	25.5181	0.0058	0.0291	17.2	17170	Accepted
24.8 mean			**25.562**	**±0.045** (combined)			n = 2	
41.2	Set 1	1	26.0556	0.0726	0.3631	1.4	1377	Accepted
41.2	Set 1	2	26.0225	0.0631	0.3157	1.6	1584	Accepted
41.2	Set 1	3	26.0142	0.0592	0.2962	1.7	1688	Accepted
41.2	Set 1	4	26.0507	0.0579	0.2897	1.7	1726	Accepted
41.2	Set 1	5	26.0706	0.0541	0.2703	1.8	1850	Accepted
41.2	Set 1	6	26.0898	0.0582	0.291	1.7	1718	Accepted
41.2	Set 1	7	26.0879	0.0561	0.2806	1.8	1782	Accepted
41.2 mean			**26.0559**	**±0.0612** (combined)			n = 7	

T (°C)	Set	Trial	f (Hz)	± interpolation (Hz)	± raw bin (Hz)	Duration (s)	Samples	Status
62	Set 1	1	26.0538	0.1186	0.5931	0.8	843	Accepted
62	Set 1	2	4.3879	0.072	0.36	1.4	1389	Rejected
62	Set 1	3	3.1523	0.0314	0.1568	3.2	3188	Rejected
62	Set 2	1	26.1785	0.0684	0.342	1.5	1462	Accepted
62	Set 2	2	26.1692	0.0661	0.3303	1.5	1514	Accepted
62	Set 2	3	26.1007	0.0548	0.274	1.8	1825	Accepted
62	Set 2	4	26.0924	0.0601	0.3005	1.7	1664	Accepted
62	Set 2	5	26.0698	0.0626	0.3129	1.6	1598	Accepted
62	Set 2	6	26.1041	0.0775	0.3876	1.3	1290	Accepted
62	Set 2	7	26.1099	0.0749	0.3745	1.3	1335	Accepted
62	Set 2	8	26.145	0.0917	0.4583	1.1	1091	Accepted
62	Set 3	1	26.2155	0.0069	0.0344	14.5	14519	Accepted
62	Set 3	2	26.1519	0.0663	0.3313	1.5	1509	Accepted
62	Set 3	3	3.0992	0.1653	0.8264	0.6	605	Rejected
62 mean			**26.1264**	**±0.0696** (combined)			n = 11	
82.1	Set 1	1	26.2934	0.0568	0.2841	1.8	1760	Accepted
82.1	Set 1	2	26.2187	0.0576	0.288	1.7	1736	Accepted
82.1	Set 1	3	26.2327	0.0607	0.3036	1.6	1647	Accepted
82.1	Set 1	4	26.2981	0.0684	0.342	1.5	1462	Accepted
82.1	Set 1	5	26.1876	0.0621	0.3106	1.6	1610	Accepted
82.1	Set 1	6	26.297	0.0783	0.3915	1.3	1277	Accepted
82.1	Set 1	7	26.2766	0.0824	0.4122	1.2	1213	Accepted

T (°C)	Set	Trial	f (Hz)	± interpolation (Hz)	± raw bin (Hz)	Duration (s)	Samples	Status
82.1	Set 1	8	26.2495	0.0782	0.3909	1.3	1279	Accepted
82.1 mean			**26.2567**	**±0.0696** (combined)			n = 8	
97.9	Set 1	1	26.1251	0.0632	0.3161	1.6	1582	Accepted
97.9	Set 1	2	26.067	0.0563	0.2814	1.8	1777	Accepted
97.9	Set 1	3	26.0946	0.0627	0.3135	1.6	1595	Accepted
97.9	Set 1	4	26.1535	0.0749	0.3745	1.3	1335	Accepted
97.9	Set 1	5	26.102	0.0701	0.3506	1.4	1426	Accepted
97.9	Set 1	6	26.2478	0.0925	0.4625	1.1	1081	Accepted
97.9	Set 1	7	26.2344	0.0885	0.4425	1.1	1130	Accepted
97.9	Set 1	8	26.0624	0.0581	0.2904	1.7	1722	Accepted
97.9 mean			**26.1358**	**±0.0751** (combined)			n = 8	
121.7	Set 1	1	26.0829	0.0124	0.0621	8.1	8054	Accepted
121.7	Set 1	2	26.1234	0.0567	0.2834	1.8	1764	Accepted
121.7	Set 1	3	26.2225	0.0615	0.3075	1.6	1626	Accepted
121.7	Set 1	4	26.1535	0.0649	0.3247	1.5	1540	Accepted
121.7	Set 1	5	26.092	0.0532	0.2661	1.9	1879	Accepted
121.7 mean			**26.1348**	**±0.0558** (combined)			n = 5	
138.4	Set 1	1	25.8993	0.0754	0.3771	1.3	1326	Accepted
138.4	Set 1	2	25.8634	0.0095	0.0476	10.5	10503	Accepted
138.4	Set 1	3	26.1348	0.0286	0.1429	3.5	3498	Accepted

T (°C)	Set	Trial	f (Hz)	± interpolation (Hz)	± raw bin (Hz)	Duration (s)	Samples	Status
138.4	Set 1	4	25.8868	0.087	0.4348	1.1	1150	Accepted
138.4 mean			**25.9461**	**±0.0808** (combined)			n = 4	

Table E1. Raw per-trial frequencies for the temperature experiment (L = 15 cm).

L (cm)	Set	Trial	f (Hz)	± interpolation (Hz)	± raw bin (Hz)	Duration (s)	Samples	Status
10	Set 1	1	43.8982	0.0789	0.3946	1.3	1267	Accepted
10	Set 1	2	43.7126	0.0791	0.3953	1.3	1265	Accepted
10	Set 1	3	43.9089	0.0904	0.4521	1.1	1106	Accepted
10	Set 1	4	43.0928	0.0643	0.3215	1.6	1555	Accepted
10	Set 1	5	3.0038	0.0546	0.2731	1.8	1831	Rejected
10 mean			**43.6531**	**±0.2074** (combined)			n = 4	
15	Set 1	1	22.4741	0.0062	0.0308	16.2	16238	Accepted
15 mean			**22.4741**	**±0.0087** (combined)			n = 1	
20	Set 1	1	13.4667	0.0168	0.0842	5.9	5936	Accepted
20	Set 1	2	13.2158	0.0158	0.0788	6.3	6347	Accepted
20	Set 1	3	13.3285	0.0292	0.1459	3.4	3428	Accepted
20 mean			**13.337**	**±0.0754** (combined)			n = 3	
25	Set 1	1	8.9249	0.0093	0.0465	10.8	10760	Accepted
25	Set 1	2	8.8958	0.0407	0.2035	2.5	2457	Accepted

L (cm)	Set	Trial	f (Hz)	± interpolation (Hz)	± raw bin (Hz)	Duration (s)	Samples	Status
25	Set 1	3	8.8961	0.0382	0.191	2.6	2618	Accepted
25 mean			**8.9056**	**±0.0309** (combined)			n = 3	
30	Set 1	1	6.3334	0.0396	0.1981	2.5	2524	Accepted
30	Set 1	2	6.23	0.0294	0.1471	3.4	3399	Accepted
30	Set 1	3	6.2457	0.028	0.1399	3.6	3573	Accepted
30	Set 1	4	6.2131	0.0351	0.1756	2.8	2848	Accepted
30	Set 1	5	6.2287	0.0286	0.1431	3.5	3493	Accepted
30	Set 2	1	6.2407	0.0274	0.1372	3.6	3645	Accepted
30	Set 2	2	6.2682	0.0286	0.1429	3.5	3499	Accepted
30	Set 2	3	6.251	0.0315	0.1575	3.2	3174	Accepted
30	Set 2	4	6.4174	0.0164	0.0821	6.1	6091	Accepted
30 mean			**6.2698**	**±0.0366** (combined)			n = 9	

Table E2. Raw per-trial frequencies for the length experiment (room temperature). The 3.0 Hz trial at 10 cm was rejected as tray vibration.

Appendix F. Validation of the Frequency Method and Model

F.1 Synthetic test of the extraction method

To check how accurately the method in Appendix E finds a frequency, it was run on 400 synthetic signals whose frequency was known exactly. Each signal imitated a real flick: a sinusoid between 25.8 and 26.4 Hz decaying with a time constant of 2.4 s, sampled at

1000 readings per second, with a 3 Hz component at 60% of its amplitude to imitate the tray vibration and 5% random noise.

Segment length (s)	Raw bin width (Hz)	ε used (Hz)	RMS error, largest bin only (Hz)	RMS error, with interpolation (Hz)
1.0	1.000	0.100	0.034	0.0023
1.5	0.667	0.067	0.024	0.0014
2.0	0.500	0.050	0.018	0.0010
5.0	0.200	0.020	0.007	0.0005

Table F1. Accuracy of the extraction method on synthetic signals of known frequency. ε = Δf_raw / 10 is the per-trial uncertainty used in Section 6.2.

The method recovers the true frequency to within about 0.002 Hz even with the tray component present, roughly 15 times better than reading the largest bin alone. The uncertainty ε assigned in Section 6.2 is therefore about 40 times larger than the method's own error, which makes it a conservative estimate.

F.2 Accuracy of the Rayleigh approximation

The Rayleigh formula (Equation 4) treats the beam as a single spring and mass. Its accuracy was checked against the exact solution for a cantilever carrying a tip mass, whose characteristic equation replaces the one in Appendix B:

$$1 + \cos(\beta L)\cosh(\beta L) + \gamma\,\beta L[\cos(\beta L)\sinh(\beta L) - \sin(\beta L)\cosh(\beta L)] = 0, \qquad \gamma = \frac{m_{tip}}{\rho A L}$$

Equation F1. Characteristic equation of a cantilever with a tip mass; with γ = 0 it reduces to Equation B1.

At L = 15 cm, γ = 1.225 g / 22.69 g = 0.054, and solving Equation F1 numerically gives $\beta_1 L = 1.785$. The Rayleigh formula predicts a frequency 1.0% higher than this exact solution at 15 cm, and 0.8 to 1.2% higher across 10 to 30 cm. Any E_0 found by inverting

Equation 4 is therefore about 2% too low, small compared with its other uncertainties. The exact solution also gives a length exponent of -1.918, almost identical to the Rayleigh value of -1.915, so the gap to the measured -1.767 in Section 7 is not caused by the approximation.

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